

CSE 311:

Data Communication

Instructor:

Dr. Md. Monirul Islam

Sampling and Analog-to-Digital Conversion

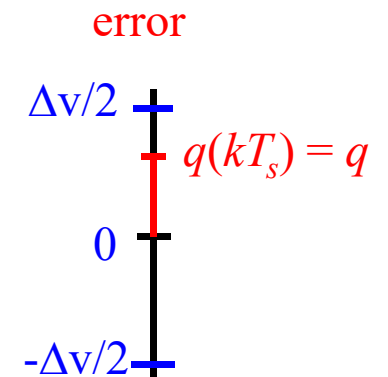
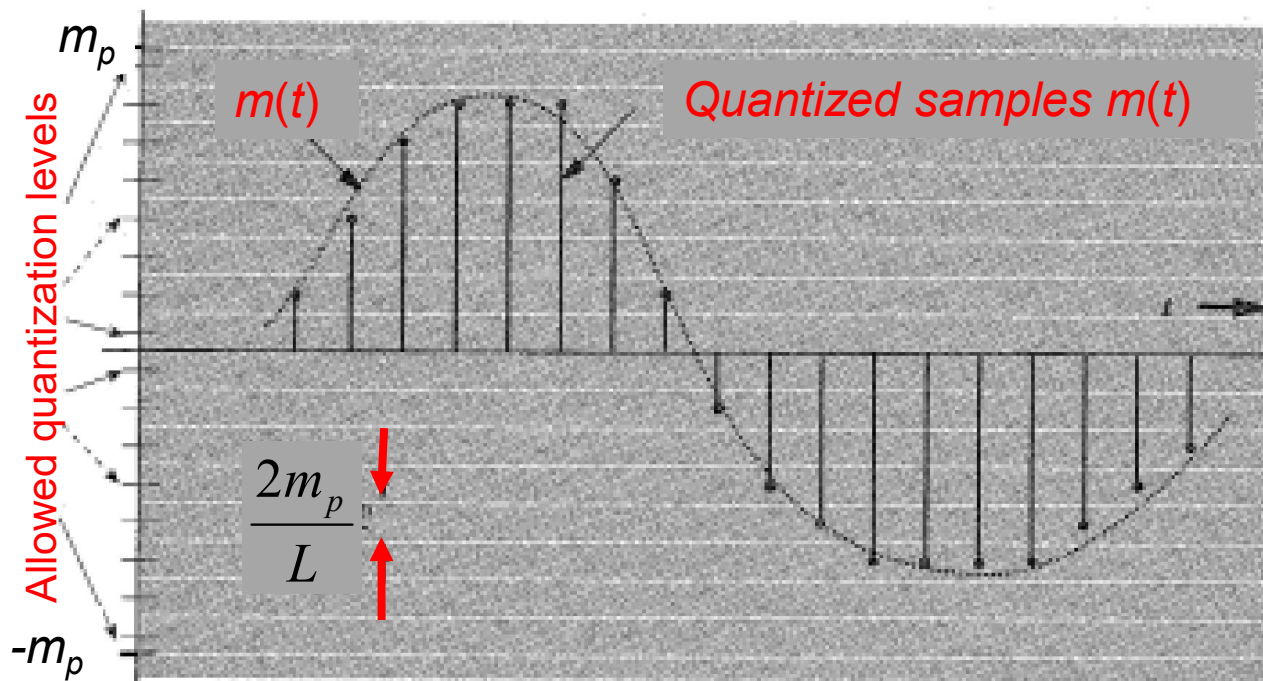
Quantization

k -th sample value $m(kT_s)$ is replaced by the midpoint of an interval where it lies

$$m(kT_s) \xrightarrow{\text{Replaced by}} \hat{m}(kT_s)$$

quantization error is unavoidable which lies in $(-\Delta v/2, \Delta v/2)$

Review



$$\Delta v = \frac{2m_p}{L}$$

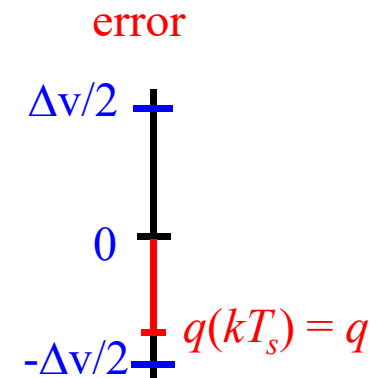
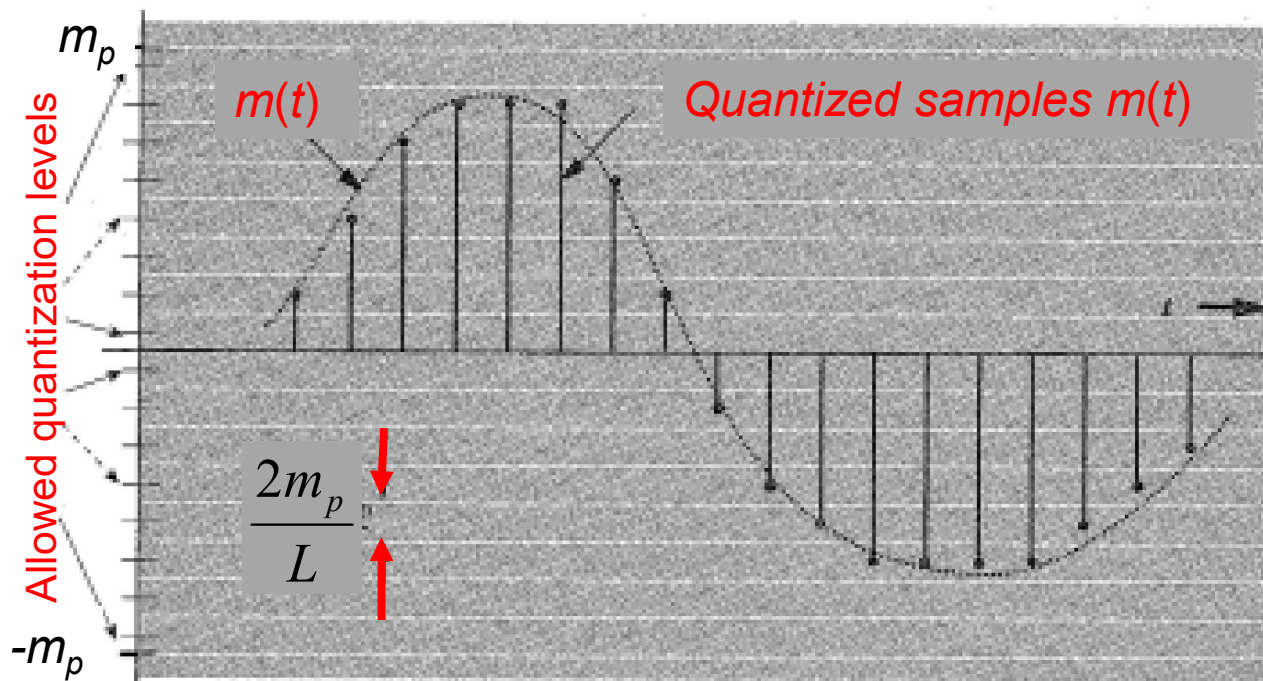
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Review



$$\Delta v = \frac{2m_p}{L}$$

Quantization

power of quantization noise (N_0) = mean square quantization error

$$N_0 = \overline{q^2(t)} = \overline{q^2} = \frac{m_p^2}{3L^2}$$

Assume, power of message signal (S_0) is given by $S_0 = \overline{m^2(t)}$

Signal-to-noise ratio (SNR) is

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

Review

Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

Review

- Higher SNR means higher quality of received signal
- L increases SNR
- Higher limit of quantizer (m_p) decreases SNR
- SNR is a linear function of signal power, $S_0 = \overline{m^2(t)}$

Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

Review

- SNR is linear function of signal power, $S_0 = \overline{m^2(t)}$

S_0 varies

- from speaker to speaker
- due to different length of connecting circuits

For these reasons,

- SNR varies widely
- Quality of received signal deteriorates remarkably for soft speakers

However, statistically,

- Small amplitudes (soft speakers) predominate in speech
- Larger amplitudes (loud speakers) are less frequent

Quantization

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

Review

Causes of the problem

- All amplitudes are **uniformly** quantized with **equal step size** $\Delta v = \frac{2m_p}{L}$
- **Variation in amplitudes** is **NOT** considered

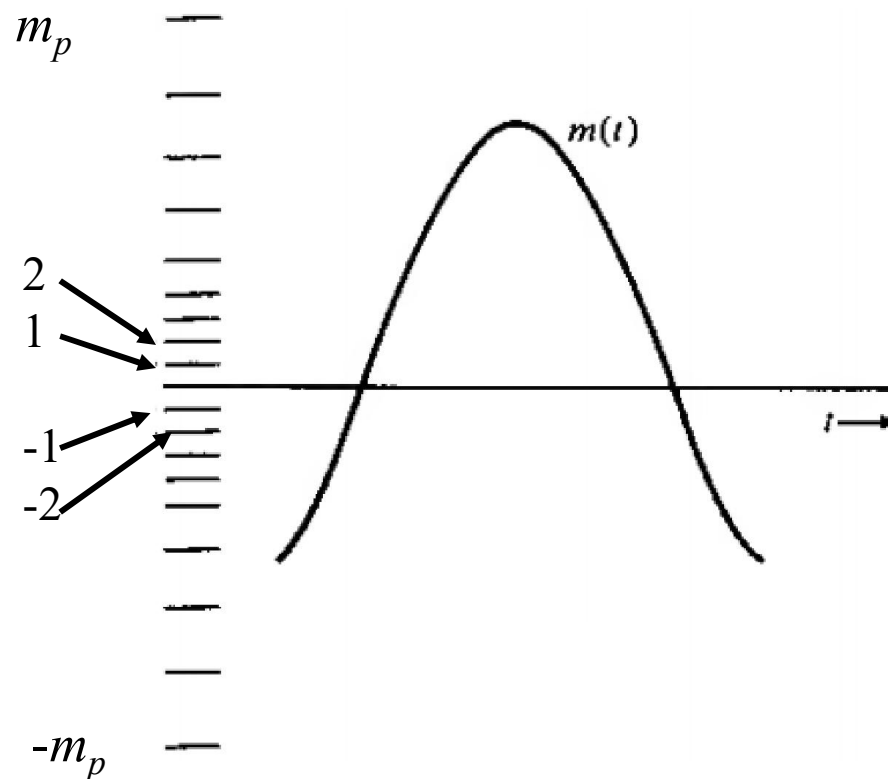
Quantization noise $N_0 = (\Delta v)^2/12$ is directly **proportional to** square of step size Δv

$$N_0 \propto (\Delta v)^2$$

Quantization

Solution to the problem: Non-uniform quantization

- Smaller step size for smaller amplitudes
- Larger step size for larger amplitudes



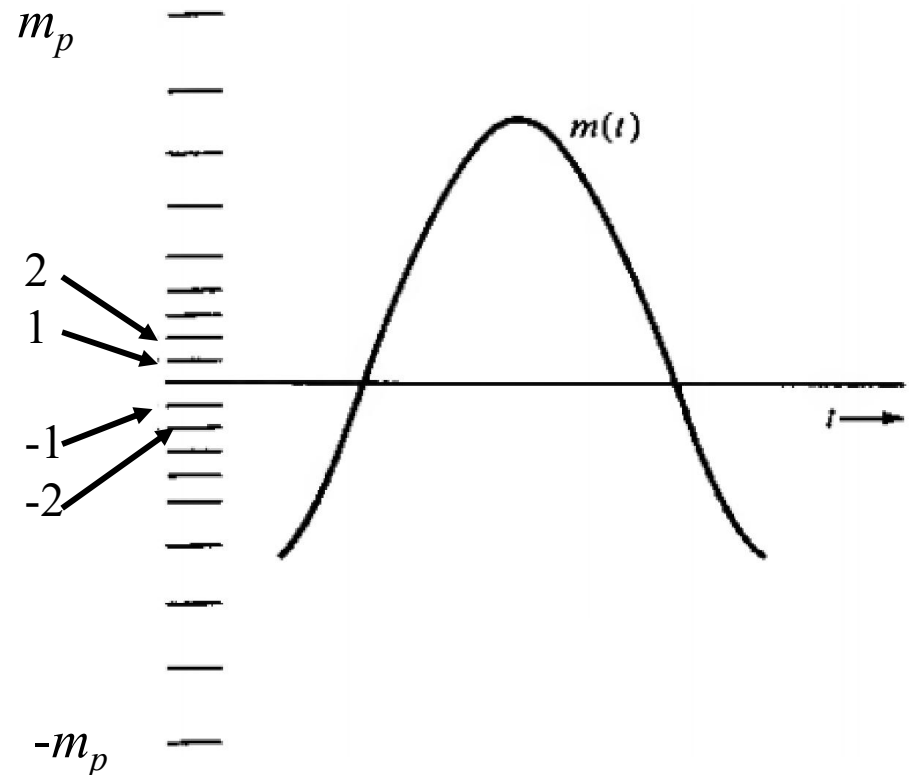
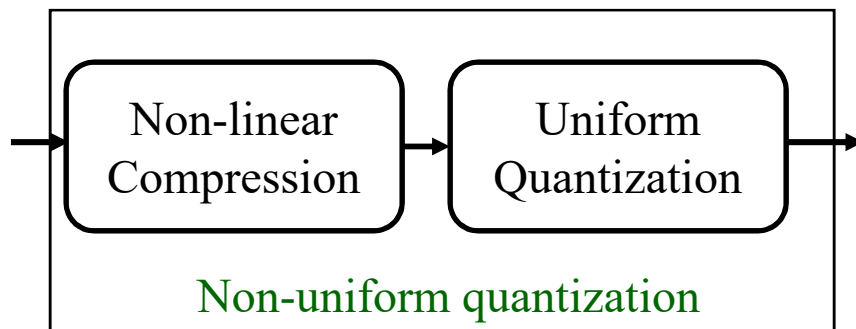
Non-uniform quantization

Quantization

Solution to the problem: Non-uniform quantization

- Smaller step size for smaller amplitudes
- Larger step size for larger amplitudes

Same result is obtained by compressing signal samples non-linearly and then using a uniform quantization

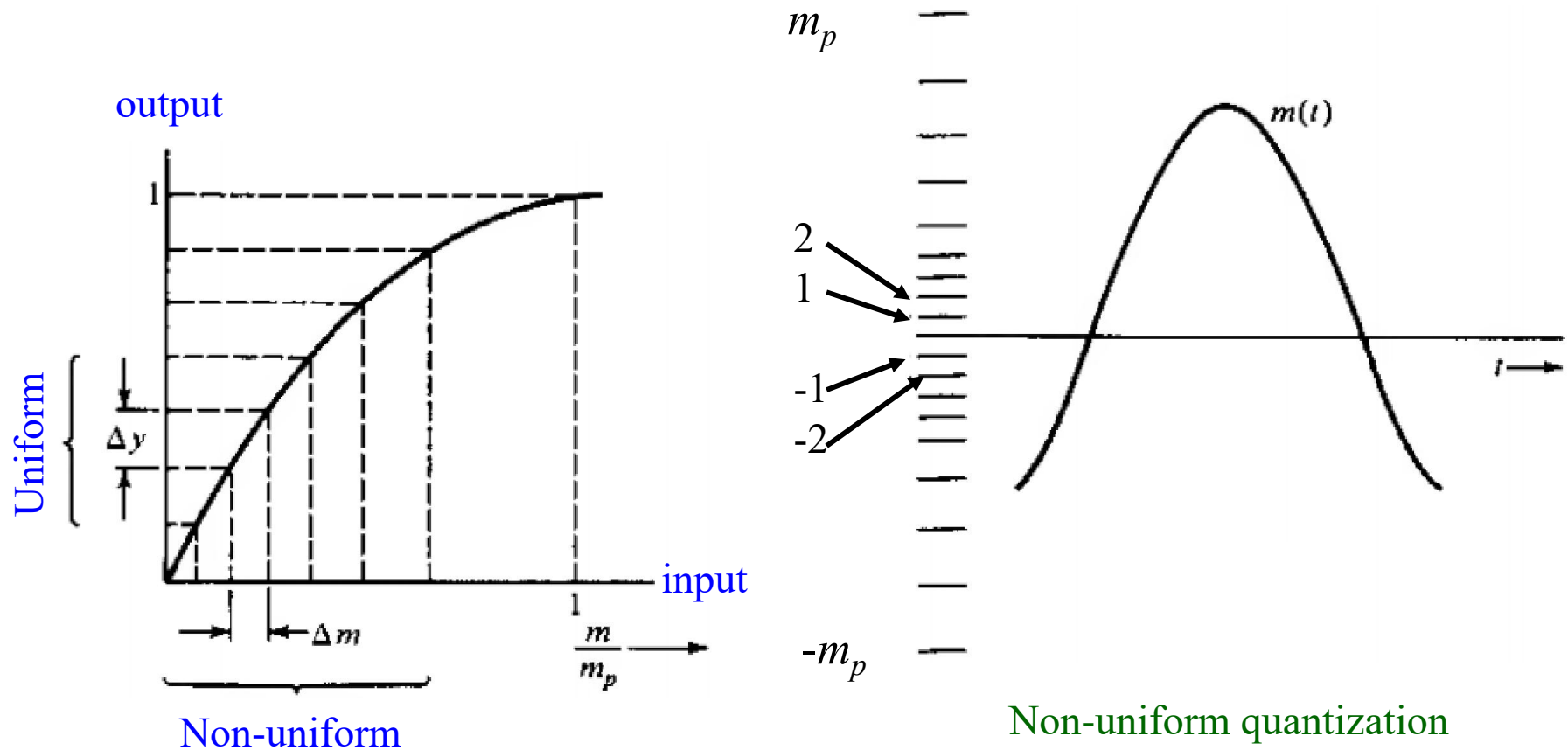


Non-uniform quantization

Quantization

Solution to the problem: Non-uniform quantization

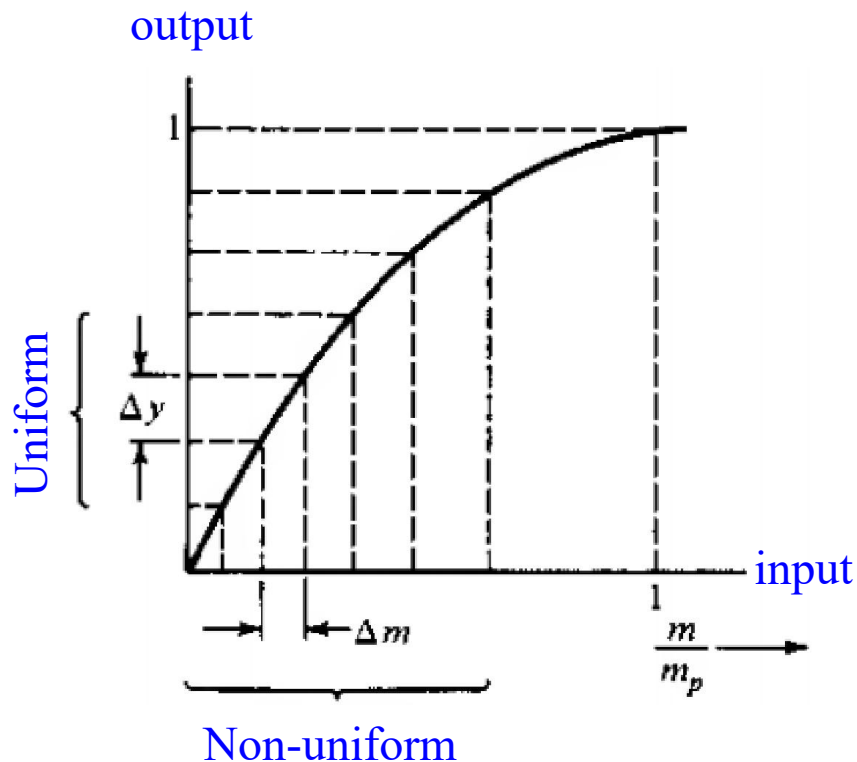
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Quantization

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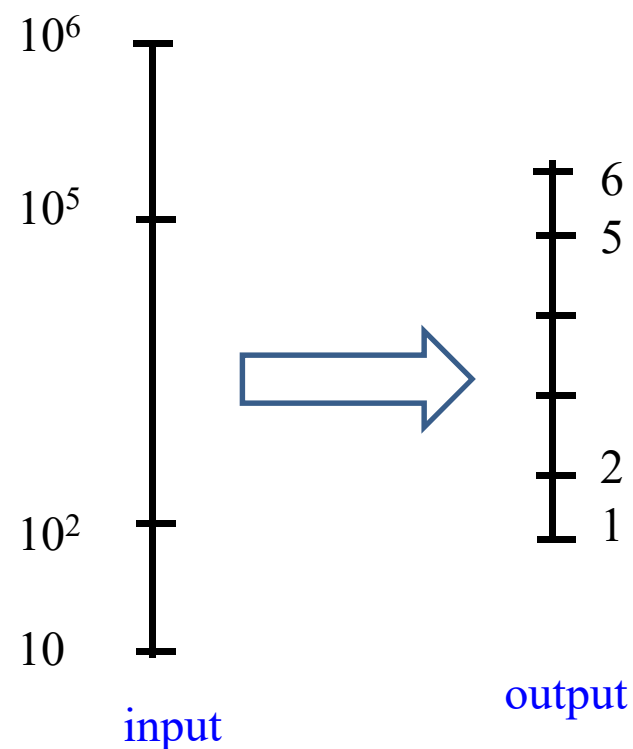
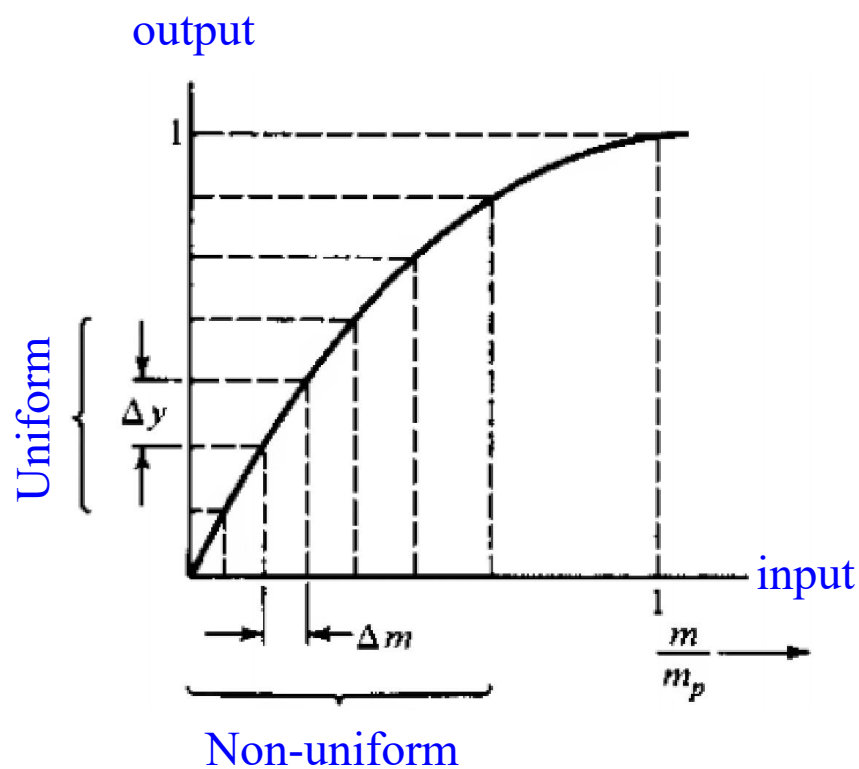


- Input signal increment Δm is mapped to larger output increment Δy for smaller m
- The opposite happens for larger m

Quantization

Solution to the problem: Non-uniform quantization

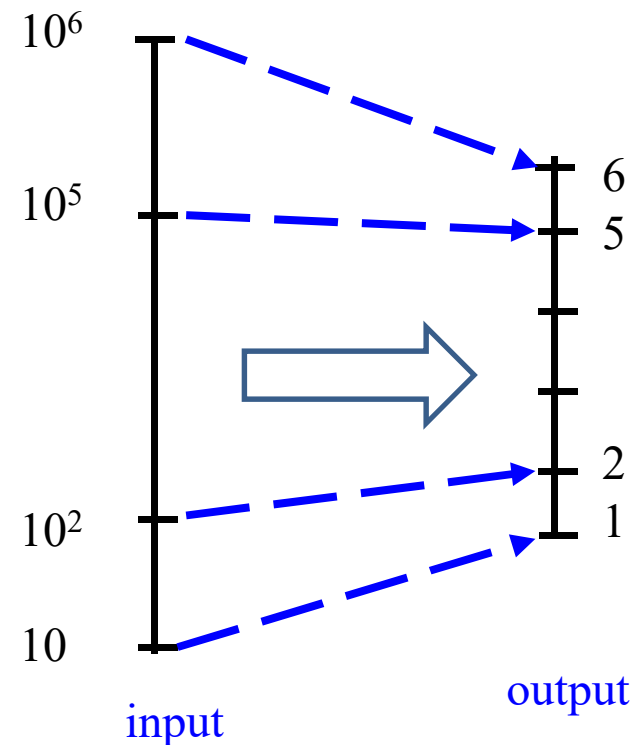
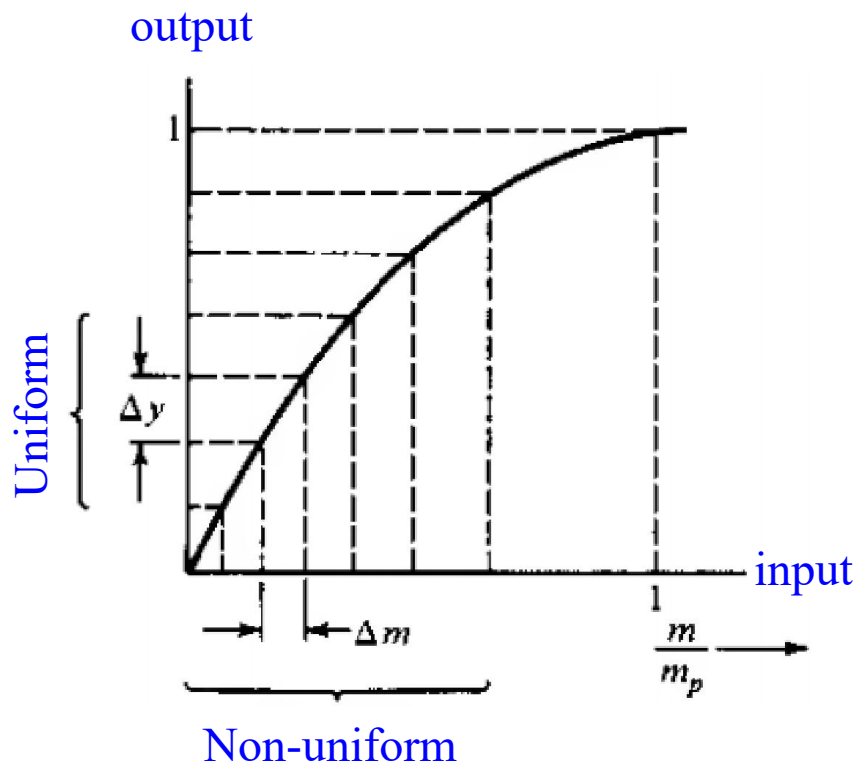
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Quantization

Solution to the problem: Non-uniform quantization

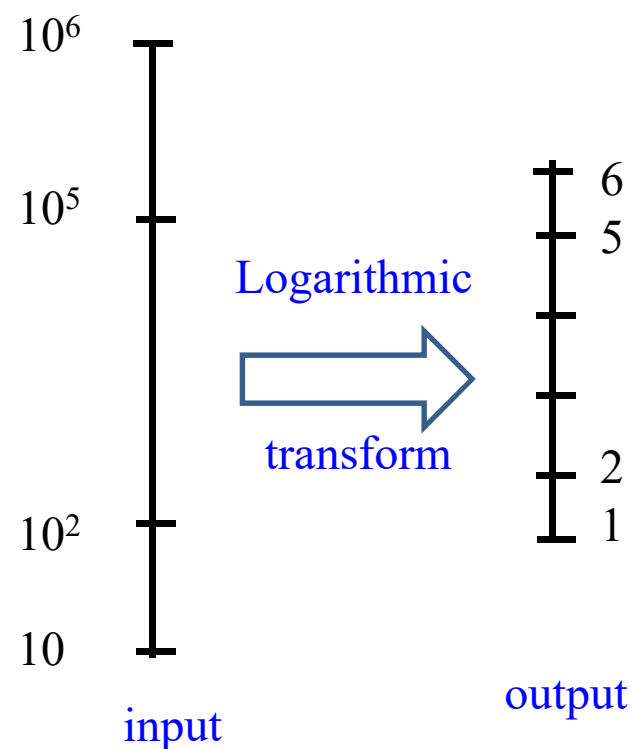
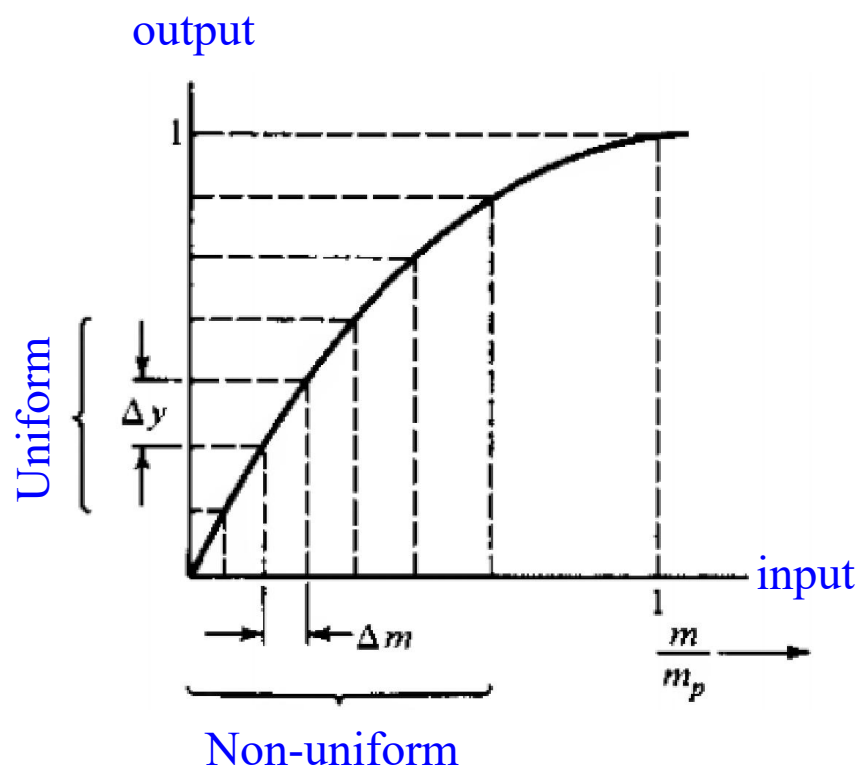
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Quantization

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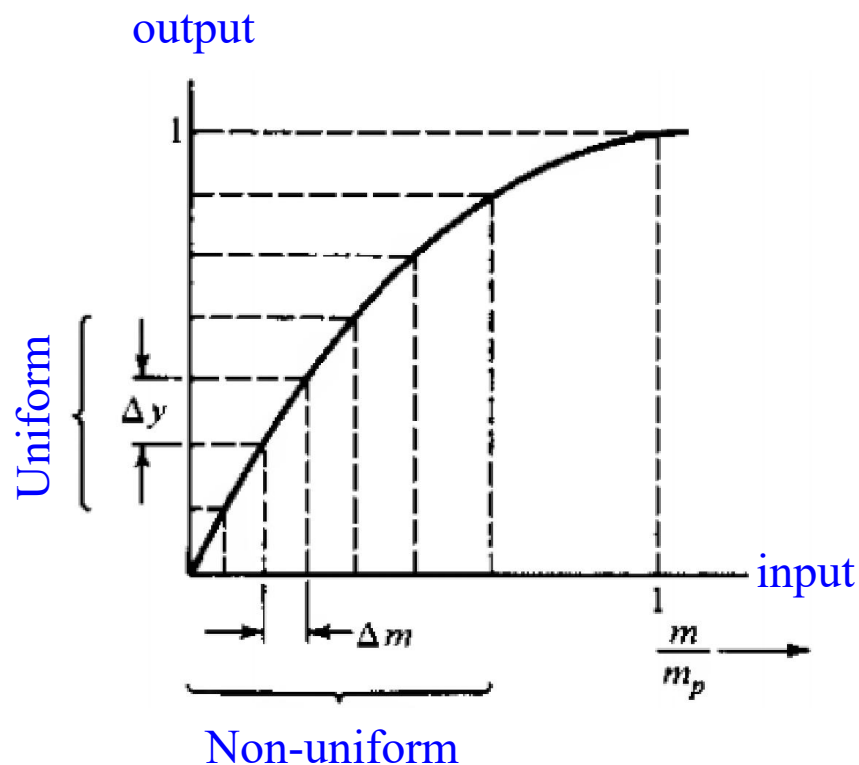
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Quantization

Solution to the problem: Non-uniform quantization

- Smaller step size for smaller amplitudes
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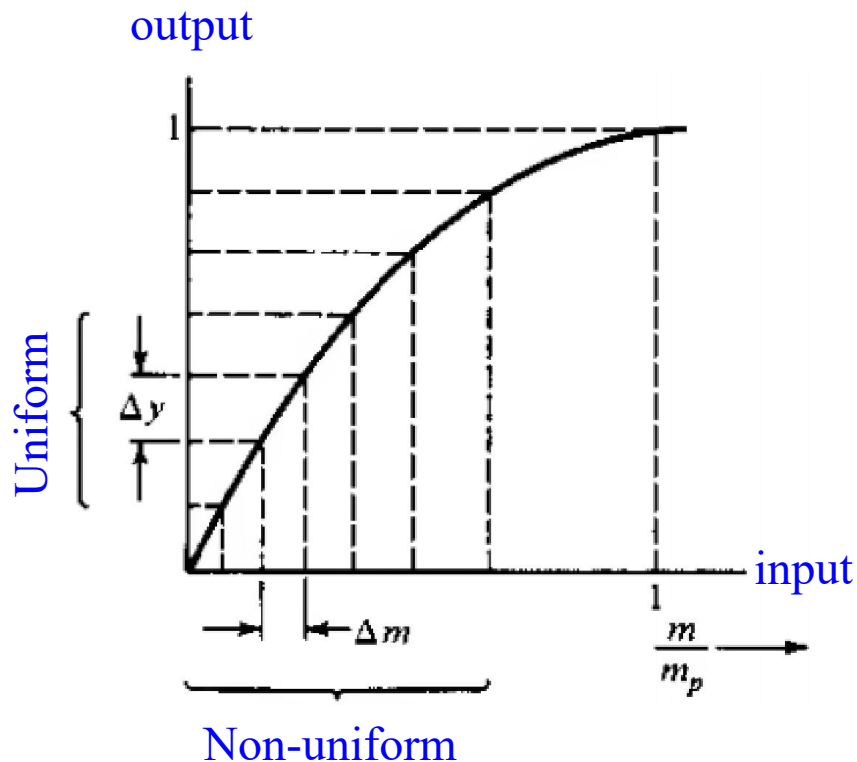


- Loud talkers are penalized by larger step size Δv to compensate soft talkers

Quantization

Solution to the problem: Non-uniform quantization

- Smaller step size for smaller amplitudes
- Larger step size for larger amplitudes



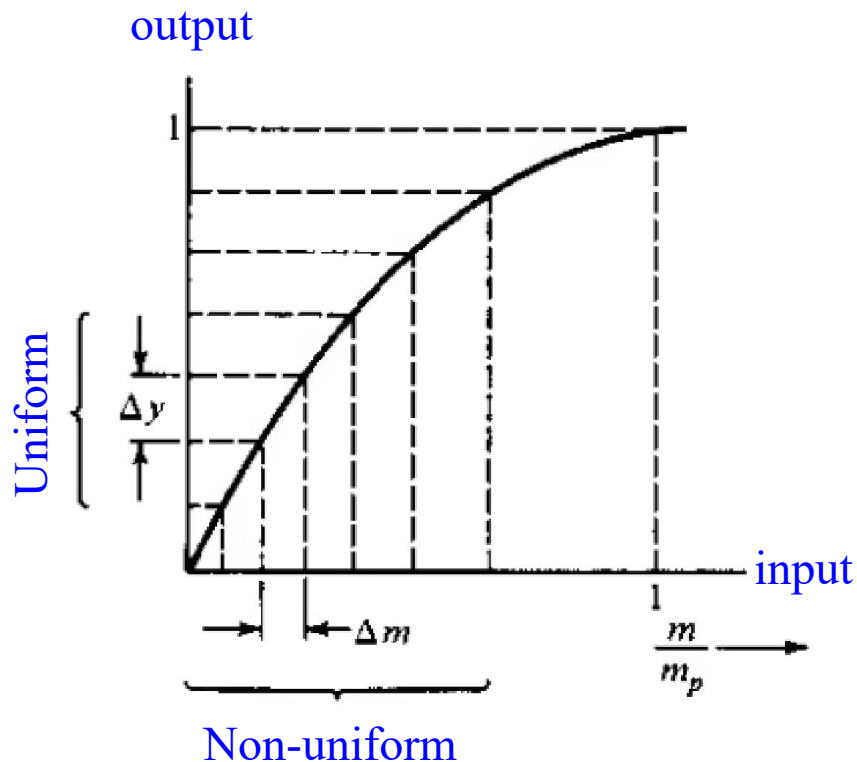
- Quantization noise becomes lower for smaller input signal $m(t)$
- The opposite happens for larger $m(t)$

$$N_0 \propto (\Delta v)^2$$

Quantization

Solution to the problem: Non-uniform quantization

- Smaller step size for smaller amplitudes
- Larger step size for larger amplitudes



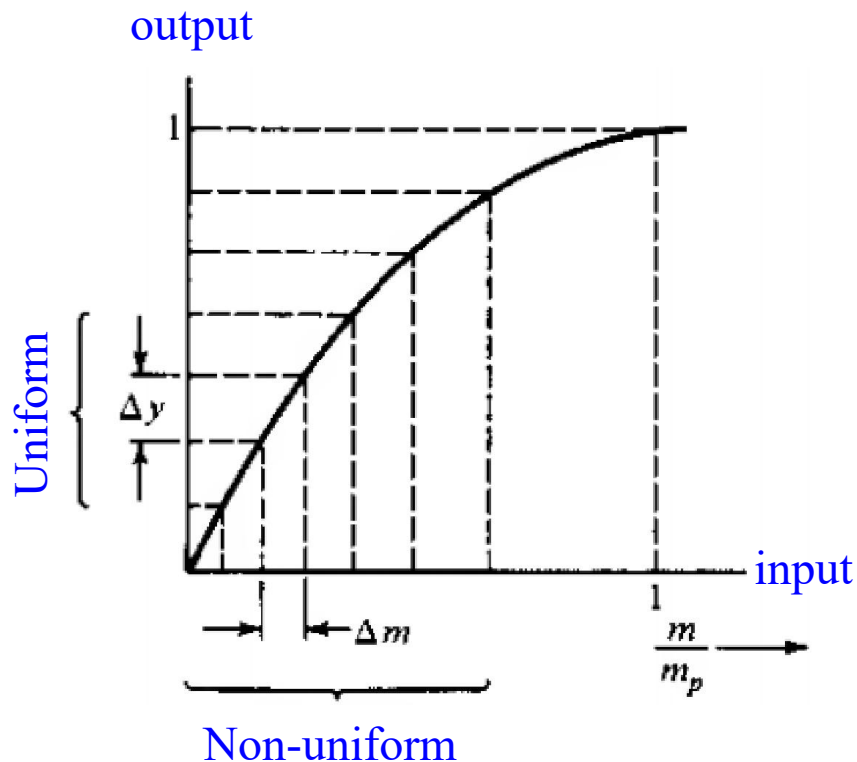
- Logarithmic compression makes quantization noise proportional to signal power

$$N_0 \propto \overline{m^2(t)} = S_0$$

Quantization

Solution to the problem: Non-uniform quantization

- Smaller step size for smaller amplitudes
- Larger step size for larger amplitudes



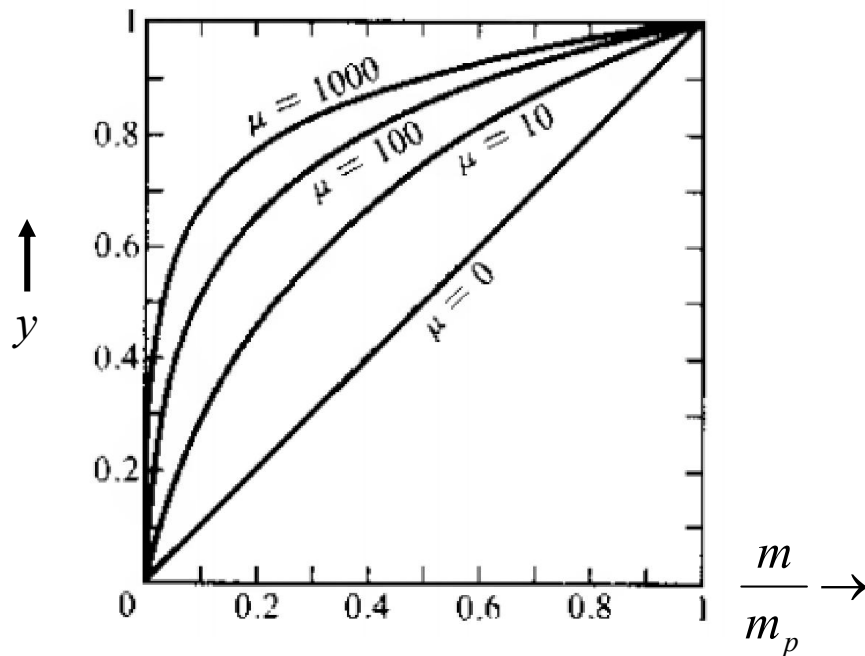
- Logarithmic compression makes quantization noise proportional to signal power

$$N_0 \propto \overline{m^2(t)} = S_0$$

$$SNR = \frac{S_0}{N_0} = \text{constant}$$

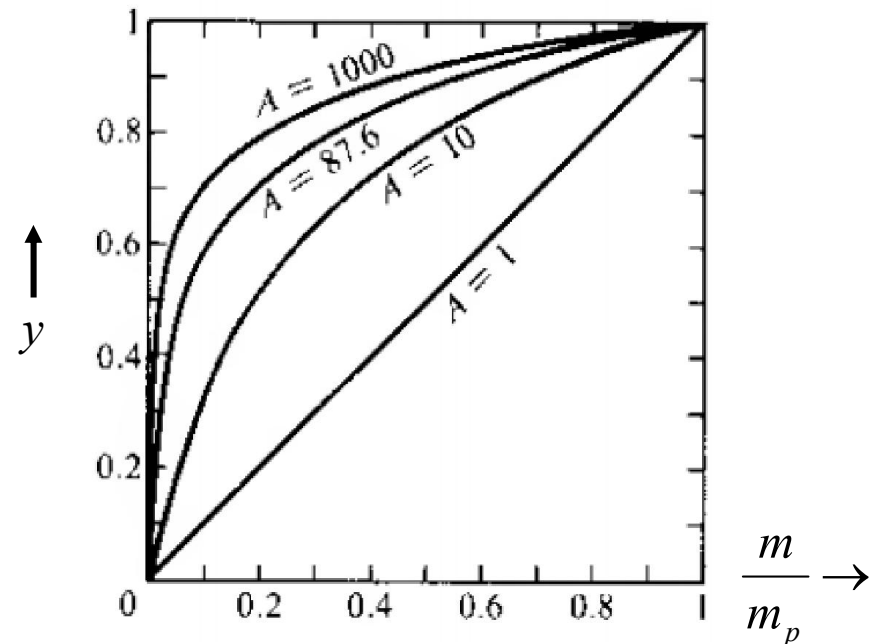
Quantization: standard Compression Characteristic

μ -law characteristic



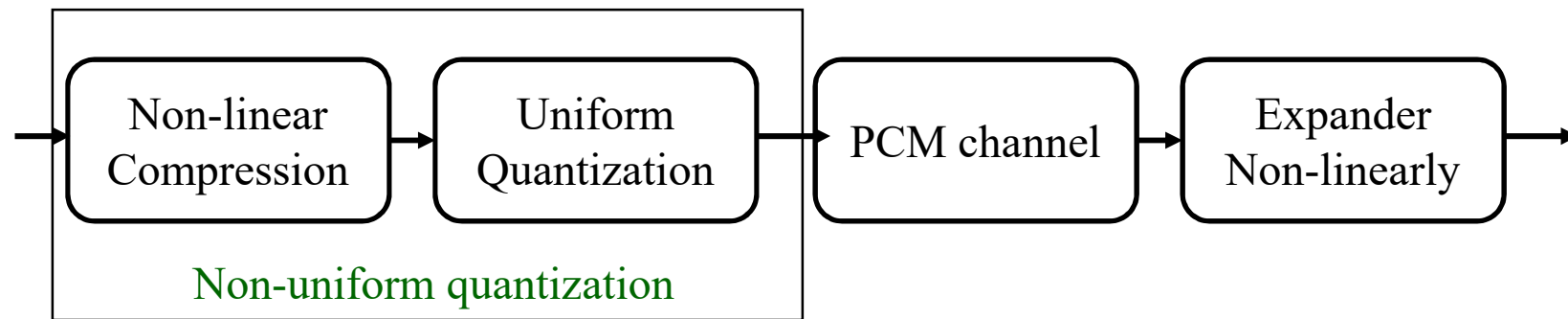
$$y = \frac{1}{\ln(1+\mu)} \ln \left(1 + \mu \frac{m}{m_p} \right) \quad 0 \leq \frac{m}{m_p} \leq 1$$

A-law characteristic



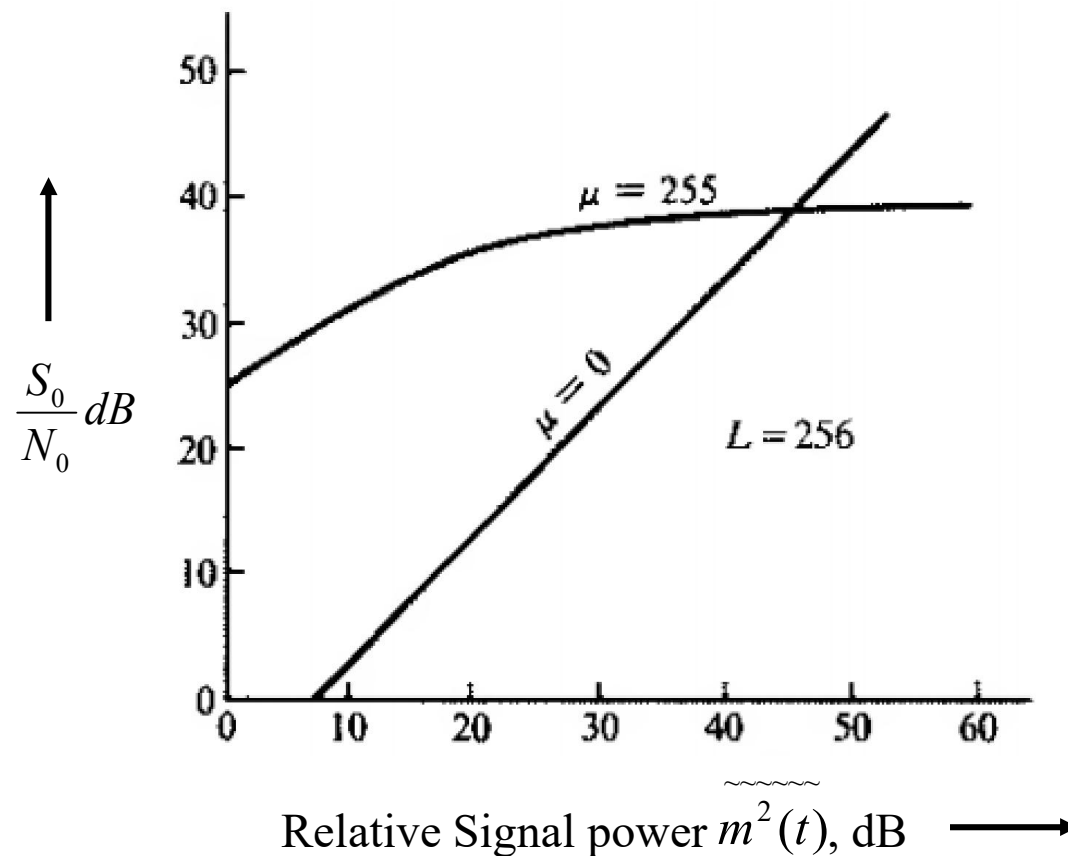
$$y = \begin{cases} \frac{A}{1 + \ln A} \left(\frac{m}{m_p} \right) & 0 \leq \frac{m}{m_p} \leq \frac{1}{A} \\ \frac{1}{1 + \ln A} \left(1 + \ln \frac{Am}{m_p} \right) & \frac{1}{A} \leq \frac{m}{m_p} \leq 1 \end{cases}$$

Restoring the Compressed Signal



Restoring the Compressed Signal

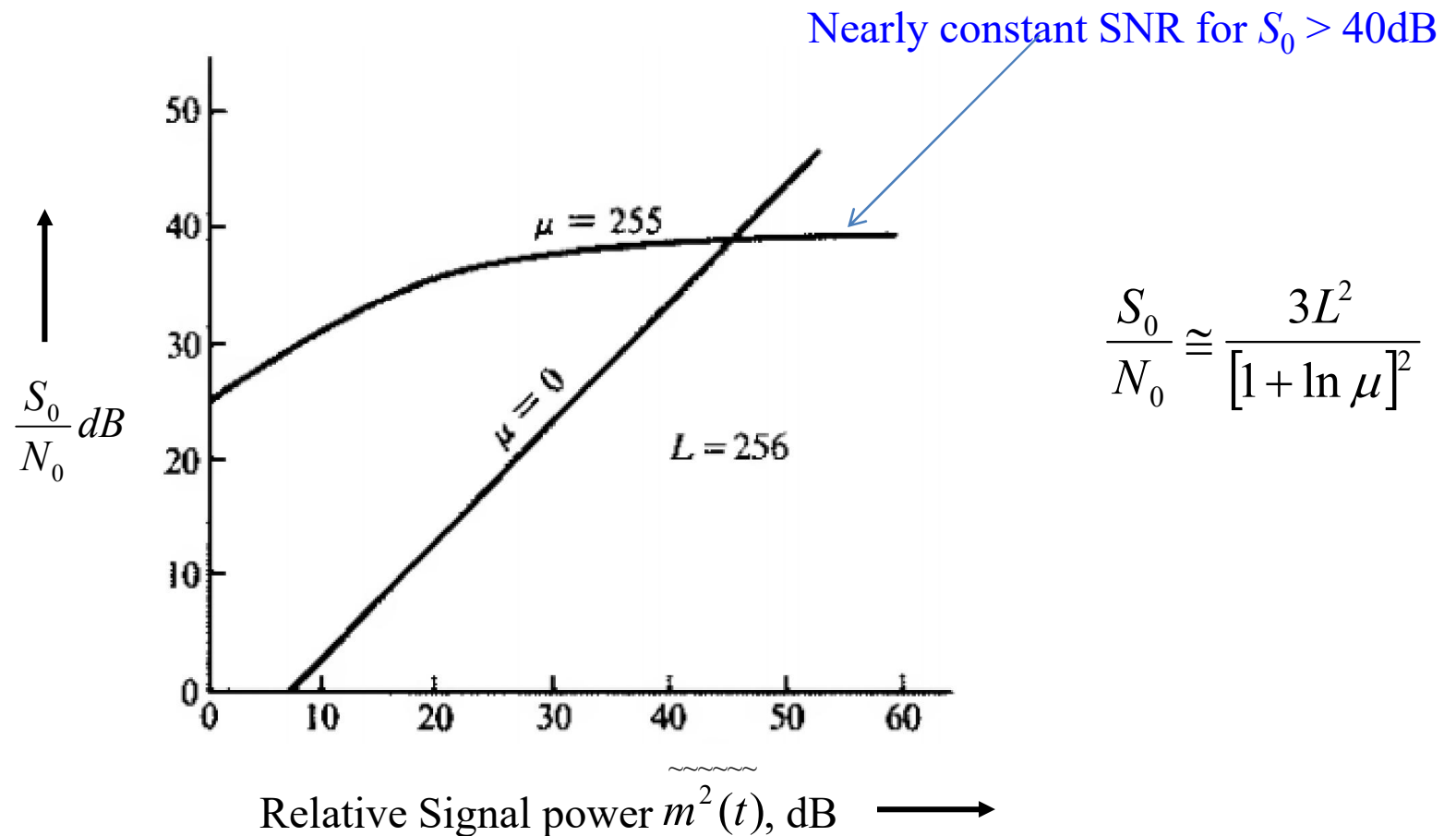
SNR improvement in μ -law compandor



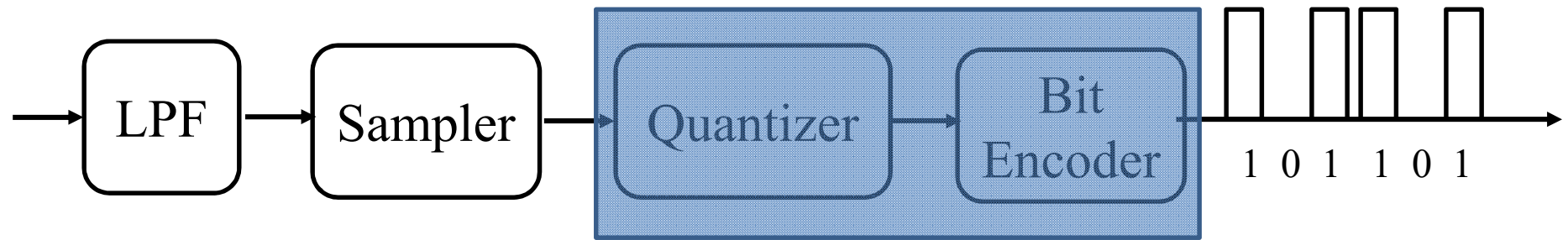
$$\frac{S_0}{N_0} \cong \frac{3L^2}{[1 + \ln \mu]^2}$$

Restoring the Compressed Signal

SNR improvement in μ -law compandor

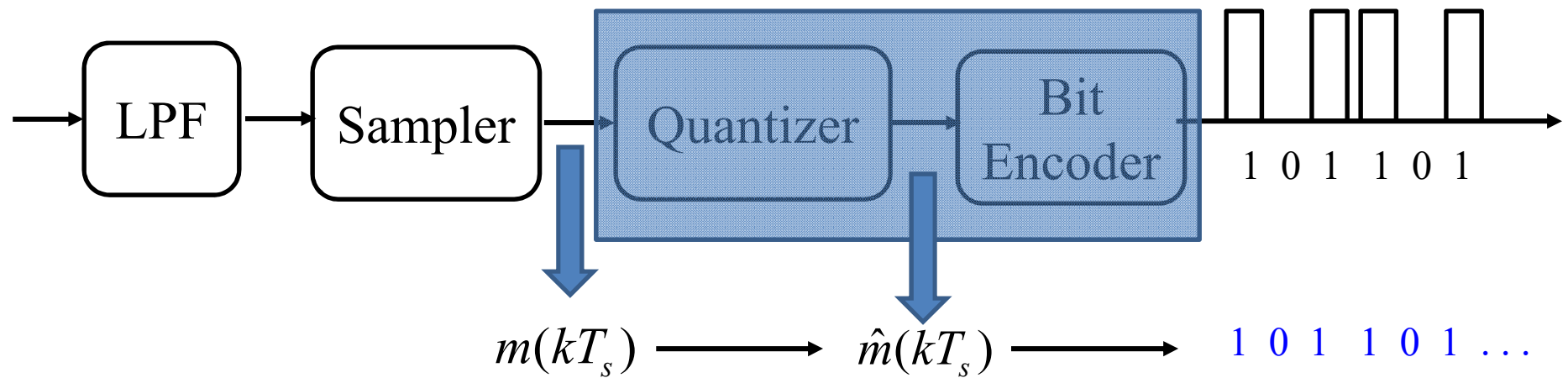


Pulse Code Modulation (PCM) Encoder

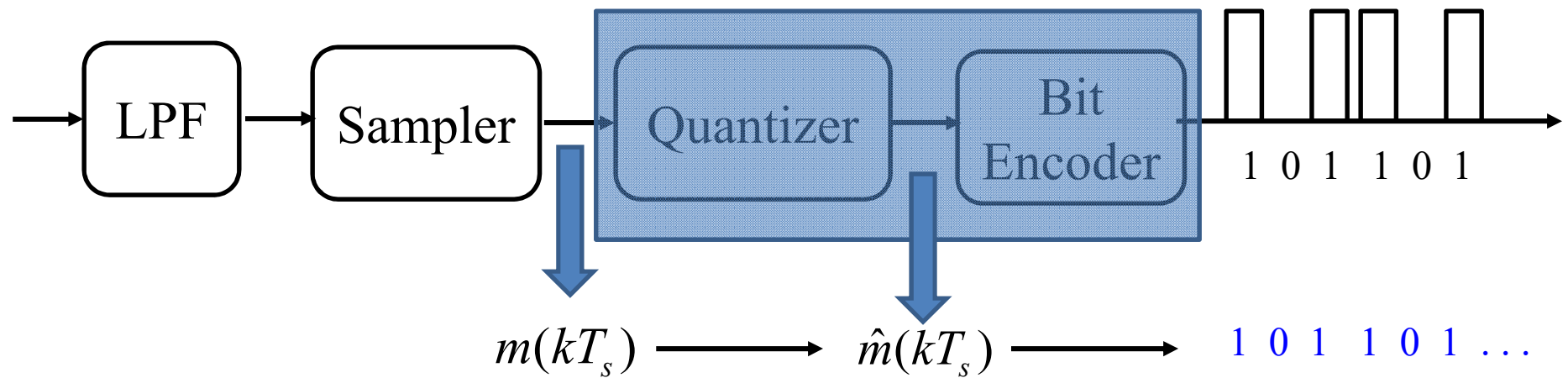


PCM system: basically an ADC

The PCM Encoder



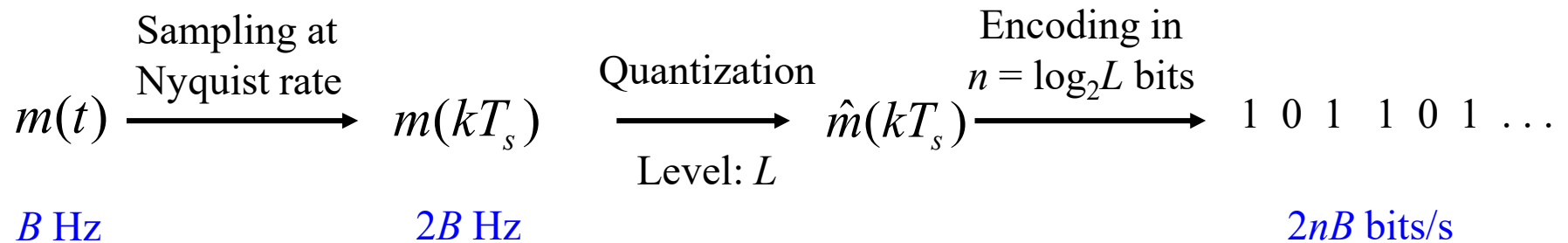
The PCM Encoder



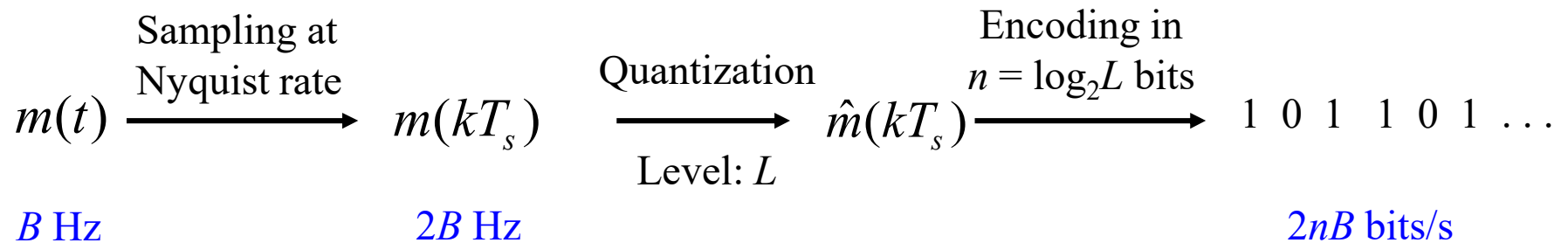
Analog-to-digital conversion:
taught at Electronics course

Self Study

Transmission Bandwidth and Output SNR

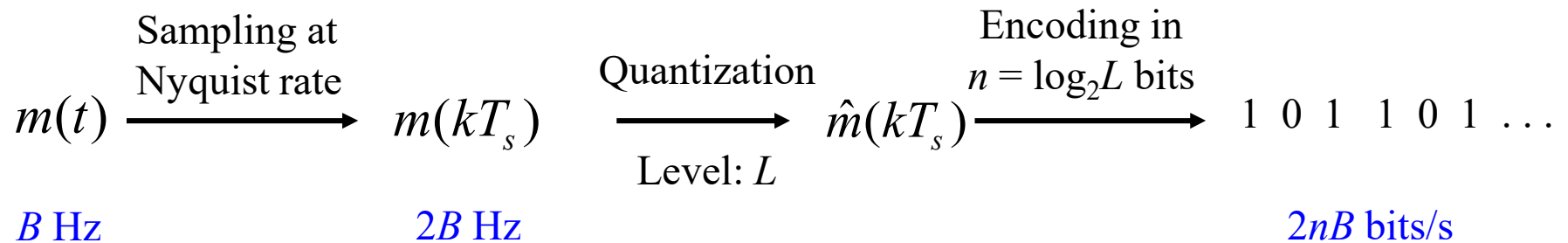


Transmission Bandwidth and Output SNR



Transmission media has to send $2nB$ bits/sec

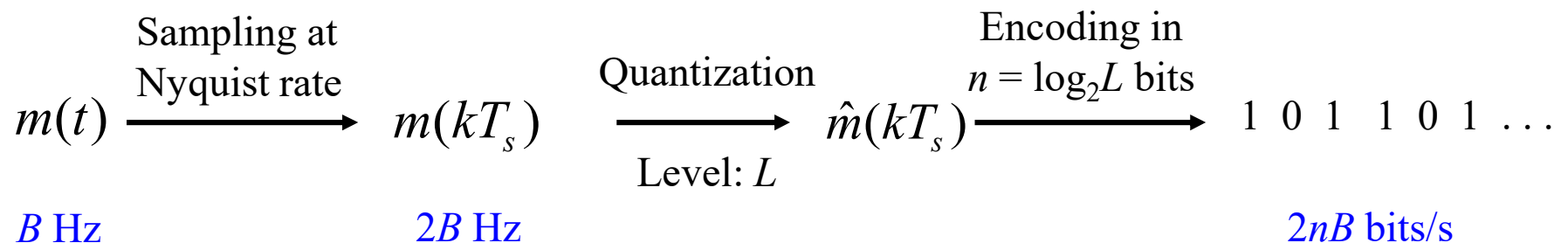
Transmission Bandwidth and Output SNR



Transmission media has to send $2nB$ bits/sec

Alternatively, transmission media has to send $2nB$ PIECES of information/sec

Transmission Bandwidth and Output SNR



Transmission media has to send $2nB$ bits/sec

Alternatively, transmission media has to send $2nB$ PIECES of information/sec

As 1 unit transmission (channel) bw (1 Hz) can transmit 2 pieces of information/sec,

Therefore, channel bandwidth, $B_T = nB$ Hz

SNR vs Channel Bandwidth, B_T

$$SNR = \frac{S_0}{N_0} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$$

$$SNR = \frac{S_0}{N_0} \cong \frac{3L^2}{[1 + \ln \mu]^2}$$

SNR vs Channel Bandwidth, B_T

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$$SNR = \frac{S_0}{N_0} \cong \frac{3L^2}{[1 + \ln \mu]^2}$$

can be combined and written as, $SNR = \frac{S_0}{N_0} = c \cdot (2)^{2n}$

SNR vs Channel Bandwidth, B_T

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can be combined and written as, $SNR = \frac{S_0}{N_0} = c \cdot (2)^{2n}$

where, $c = \begin{cases} \frac{\overline{3m^2(t)}}{m_p^2} & \text{uncompressed case} \\ \frac{3}{[1 + \ln \mu]^2} & \text{compressed case} \end{cases}$

and, $L = 2^n$

SNR vs Channel Bandwidth, B_T

$$SNR = \frac{S_0}{N_0} = c \cdot (2)^{2n}$$

Replace n by B_T/B where,

$$\frac{S_0}{N_0} = c \cdot (2)^{2B_T / B}$$

SNR exponentially increases with channel bandwidth B_T

SNR vs Channel Bandwidth, B_T

$$SNR = \frac{S_0}{N_0} = c \cdot (2)^{2n}$$

$$\left(\frac{S_0}{N_0} \right)_{dB} = 10 \log_{10} \left(\frac{S_0}{N_0} \right) = 10 \log_{10} (c \cdot (2)^{2n})$$

SNR vs Channel Bandwidth, B_T

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$$= 10 \log_{10} c + 20n \log_{10} 2$$

$$= (\alpha + 6n) \text{ dB}$$

SNR vs Channel Bandwidth, B_T

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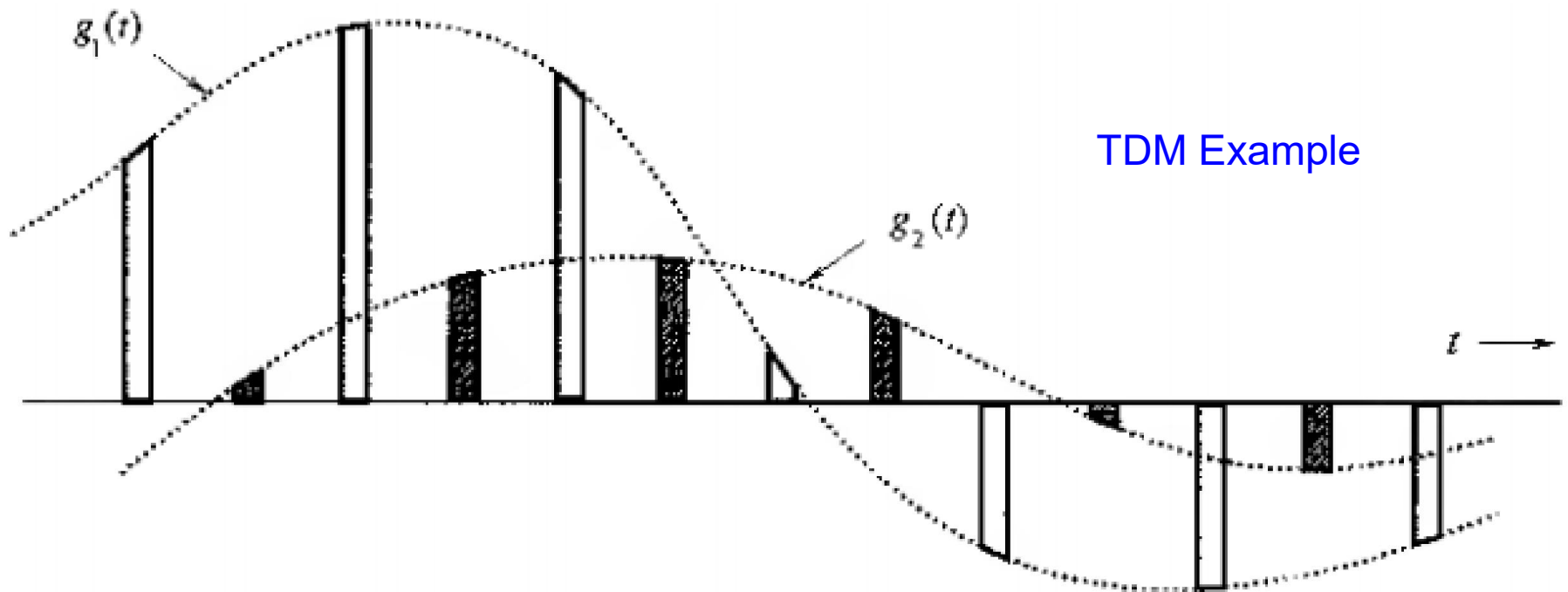
$$= 10 \log_{10} c + 20n \log_{10} 2$$

$$= (\alpha + 6n) \text{ dB}$$

1 bit increase in L quadruples SNR ($\equiv$ 6 dB increase)

Application of PCM

Time Division Multiplexing (TDM): Pulses from multiple signals are interleaved on the same channel



PCM in TDM: T1 Carrier System - A Case Study

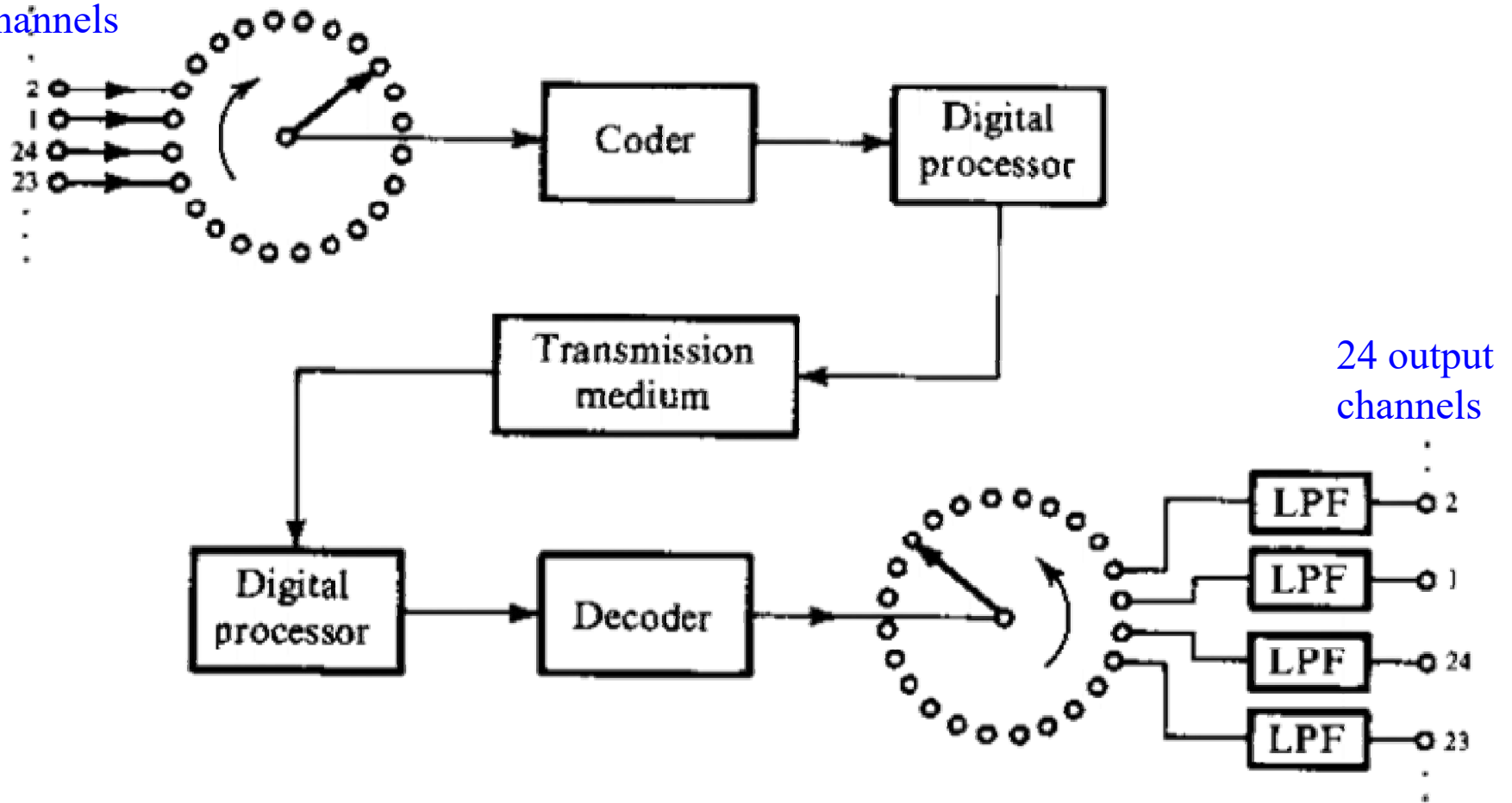
T1 Carrier

- physically a pair of copper wires
- replaces earlier technology that transmitted a single analog audio signal at a time
- now transmits 24 time-division-multiplexed PCM telephone signals
- can transmit 24 telephone calls simultaneously
- Current total transmission speed is 1.544 Mbps compared to previous 4 KHz analog bandwidth

PCM in TDM: T1 Carrier System

- A Case Study

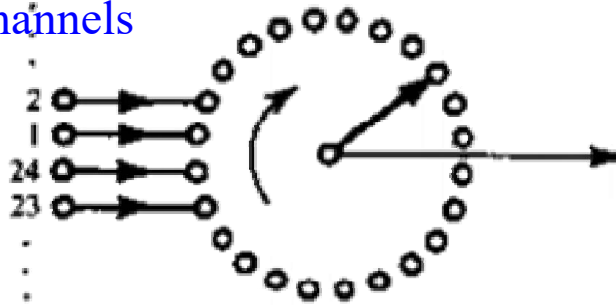
24 input
channels



PCM in TDM: T1 Carrier System

- A Case Study

24 input
channels

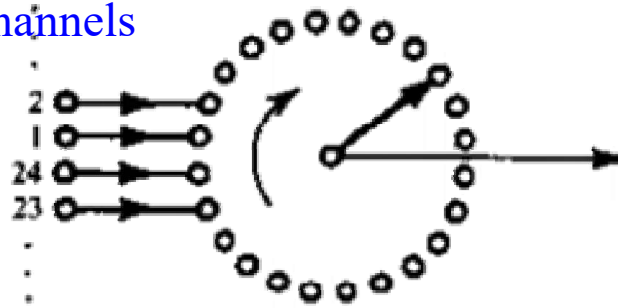


- 24 input channels each carrying voice signal of 4KHz

PCM in TDM: T1 Carrier System

- A Case Study

24 input
channels



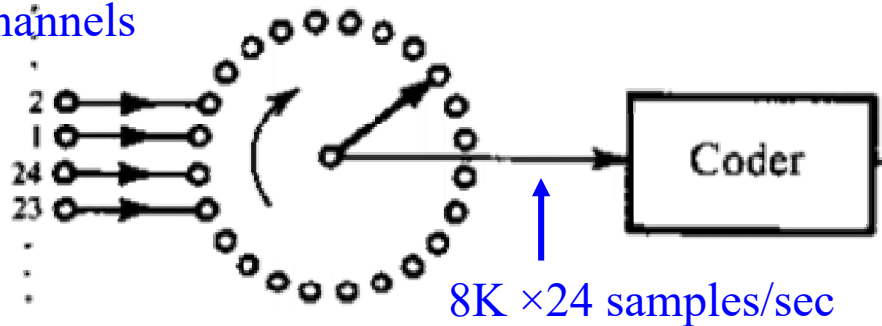
Commutators

- work as samplers
- are high speed electronic switching circuits
- are opened periodically with narrow pulses of $2 \mu\text{s}$
- sample 24 channels in turns at the rate of 8 KHz
- generates $8\text{K} \times 24$ samples/second

PCM in TDM: T1 Carrier System

- A Case Study

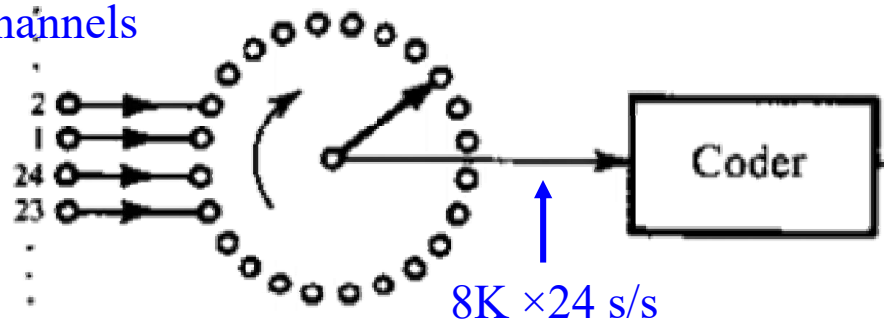
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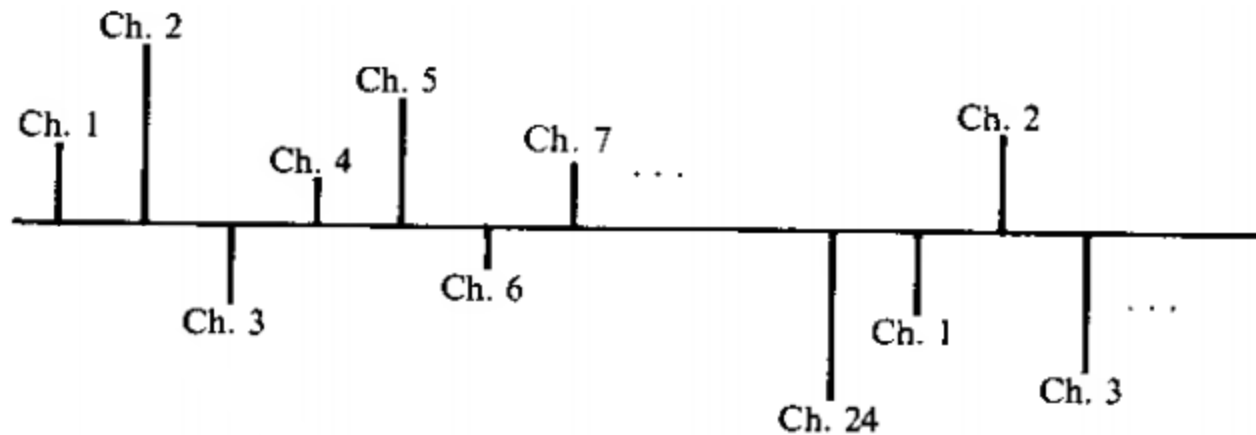
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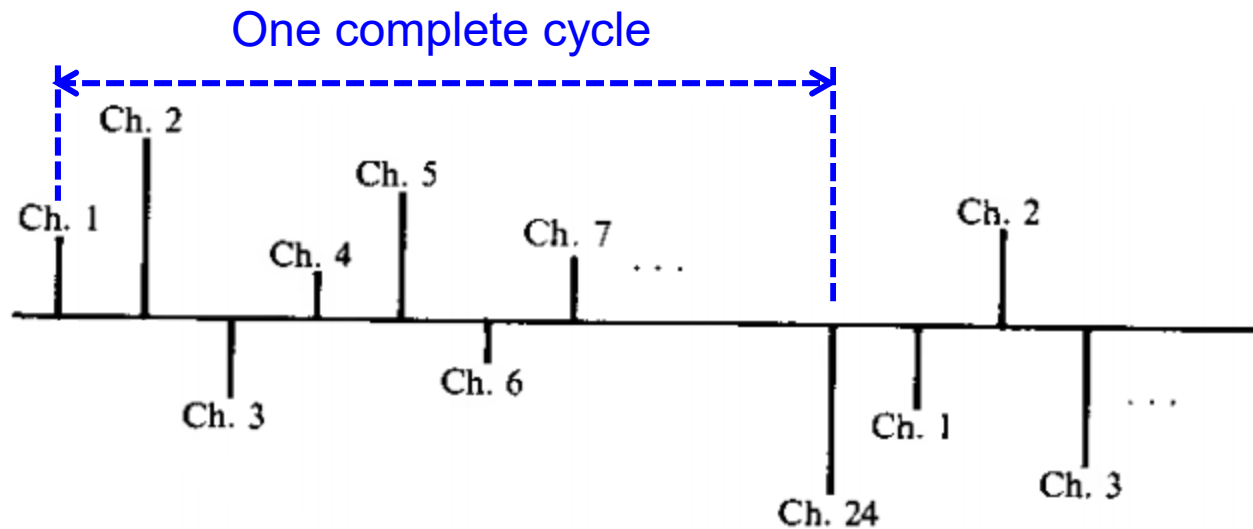
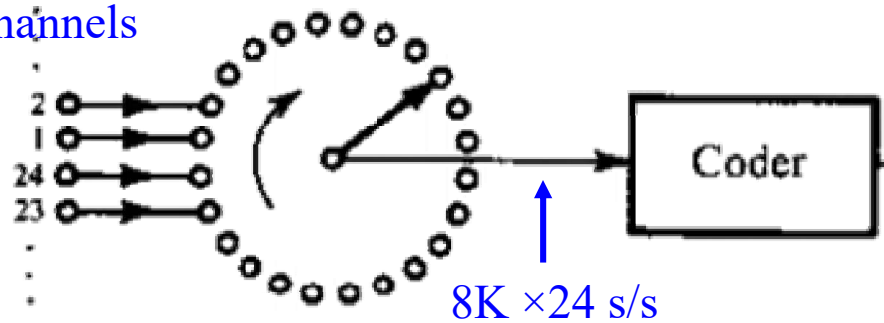
Samples from different channels



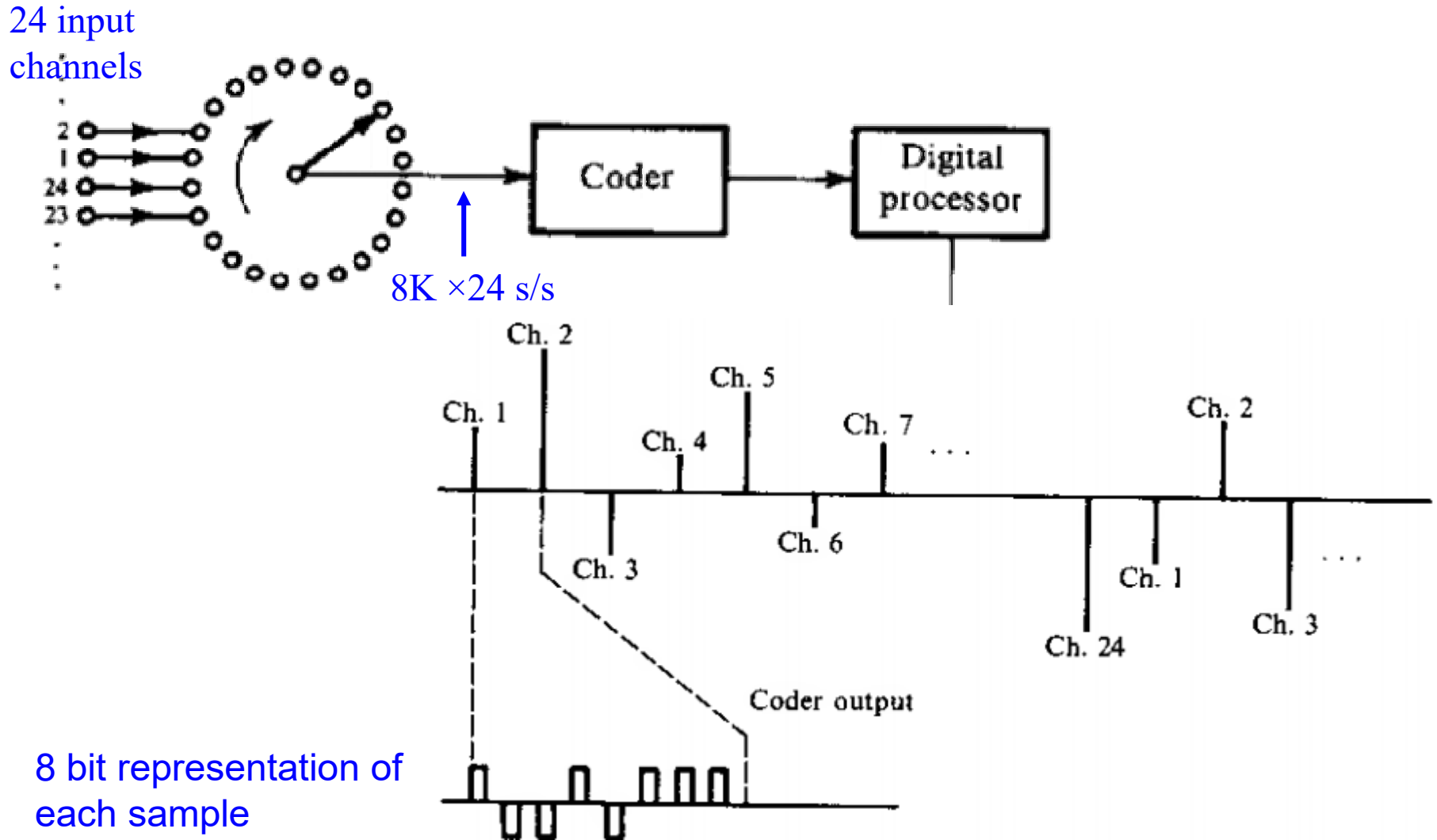
PCM in TDM: T1 Carrier System

- A Case Study

24 input
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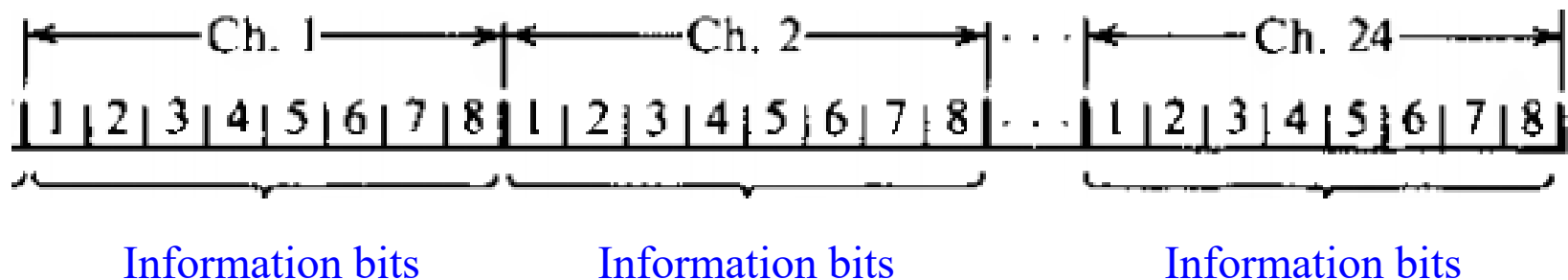
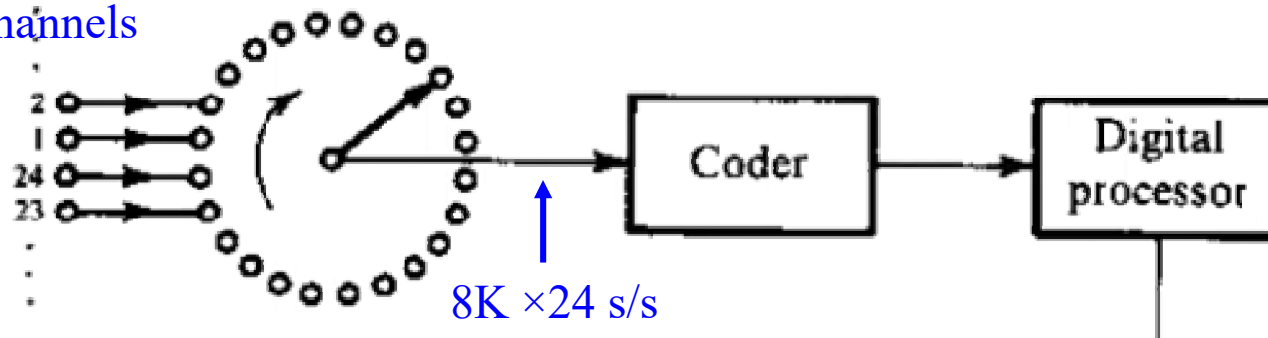
PCM in TDM: T1 Carrier System - A Case Study



PCM in TDM: T1 Carrier System

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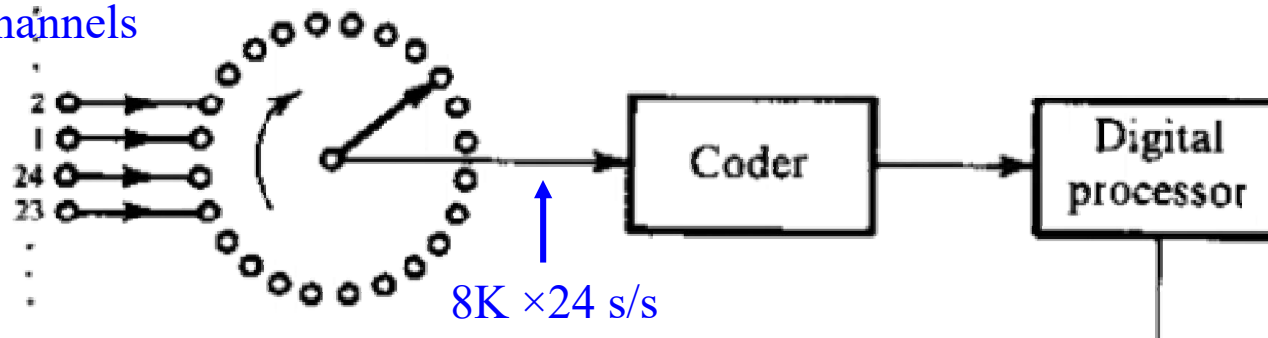
24 input
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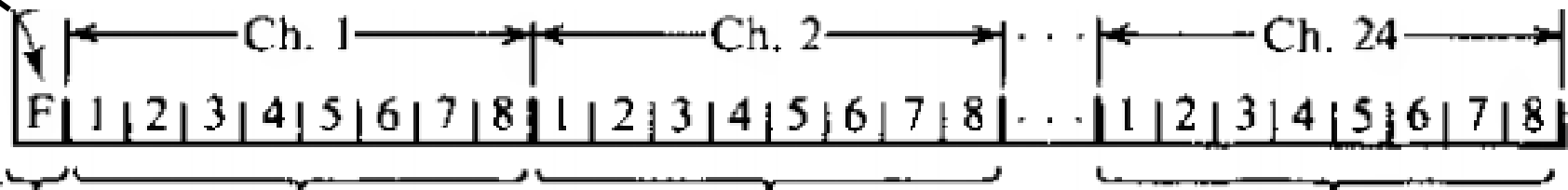
PCM in TDM: T1 Carrier System

- A Case Study

24 input
channels



Framing bit



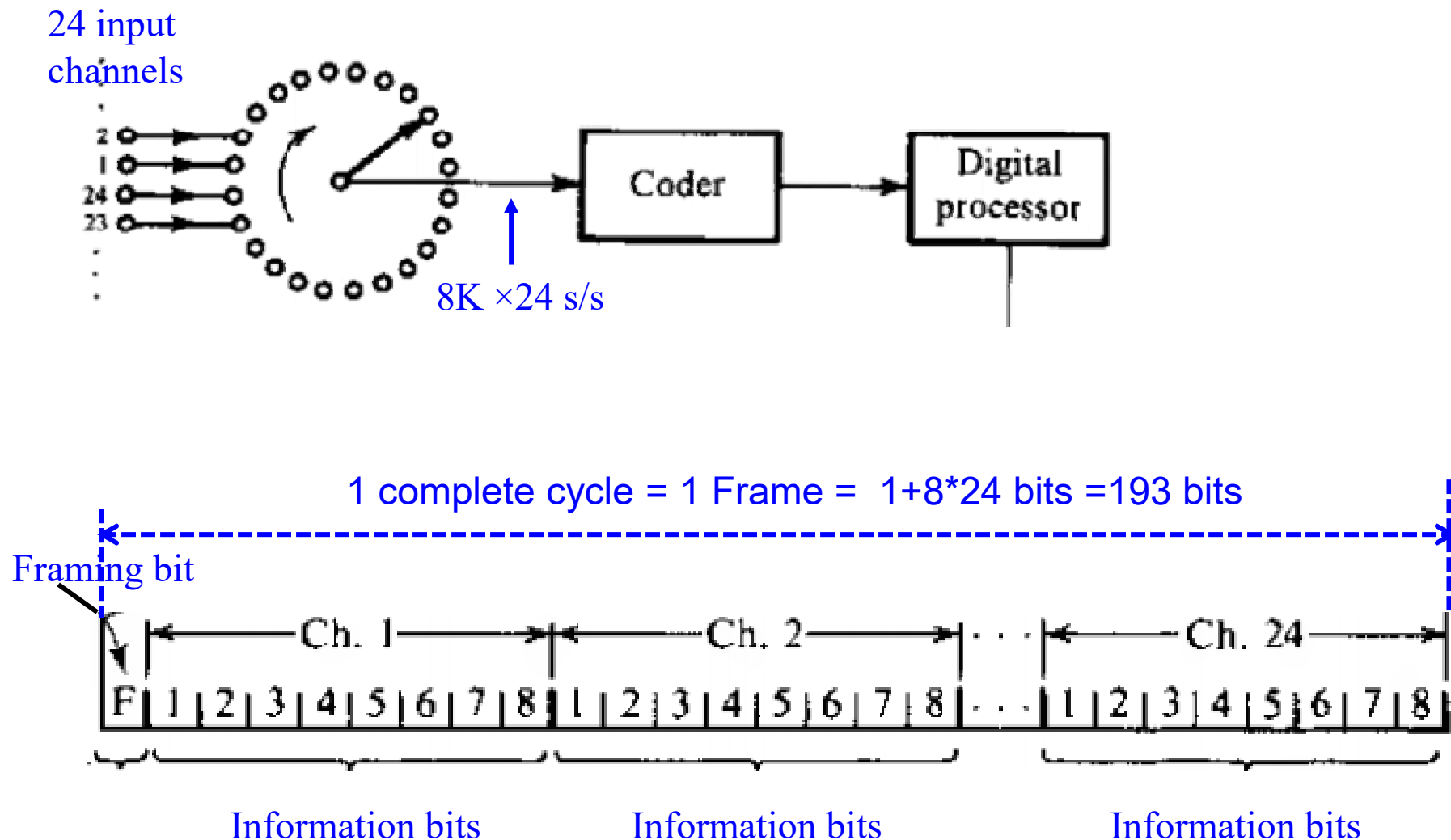
Information bits

Information bits

Information bits

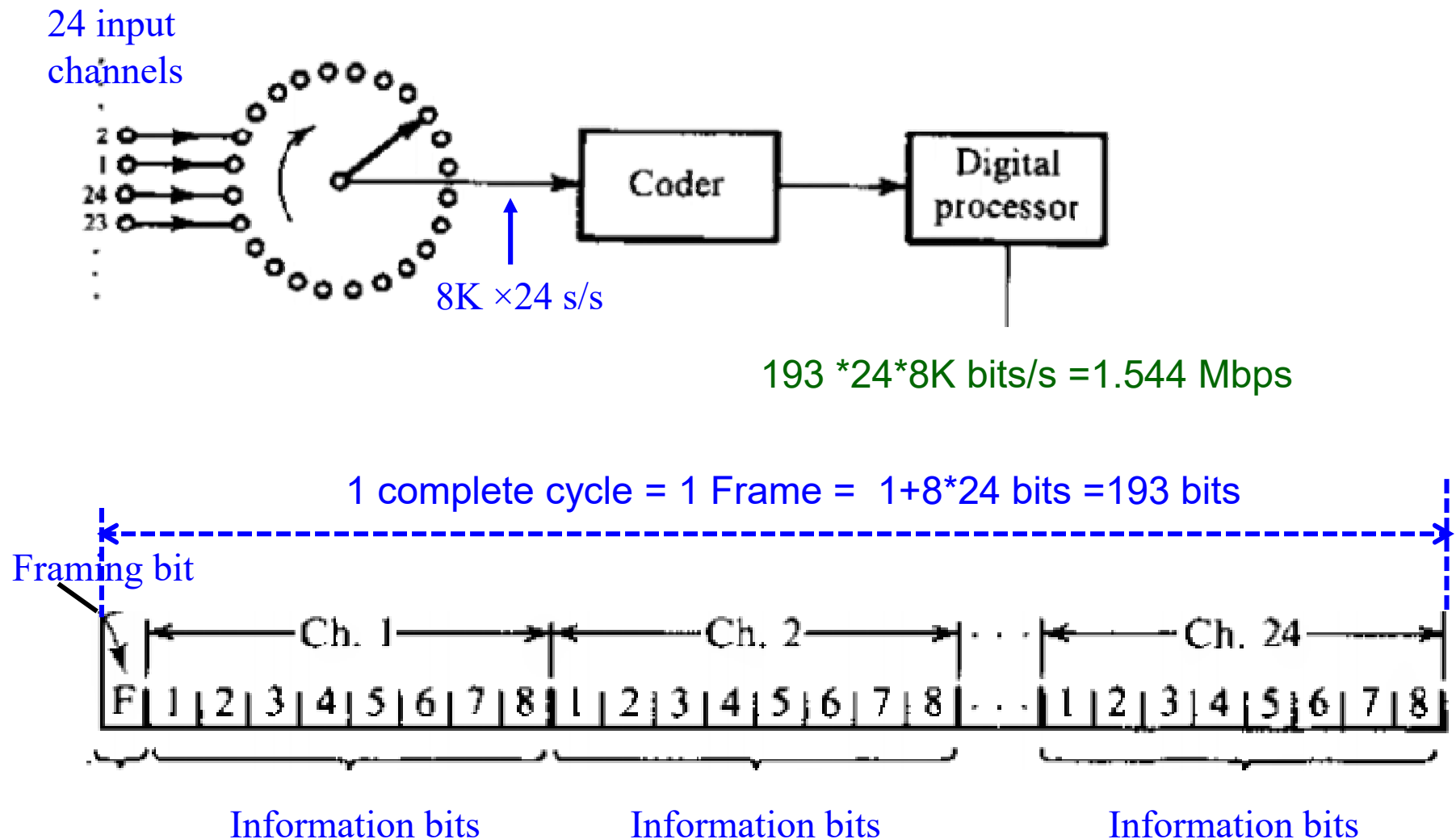
PCM in TDM: T1 Carrier System

- A Case Study



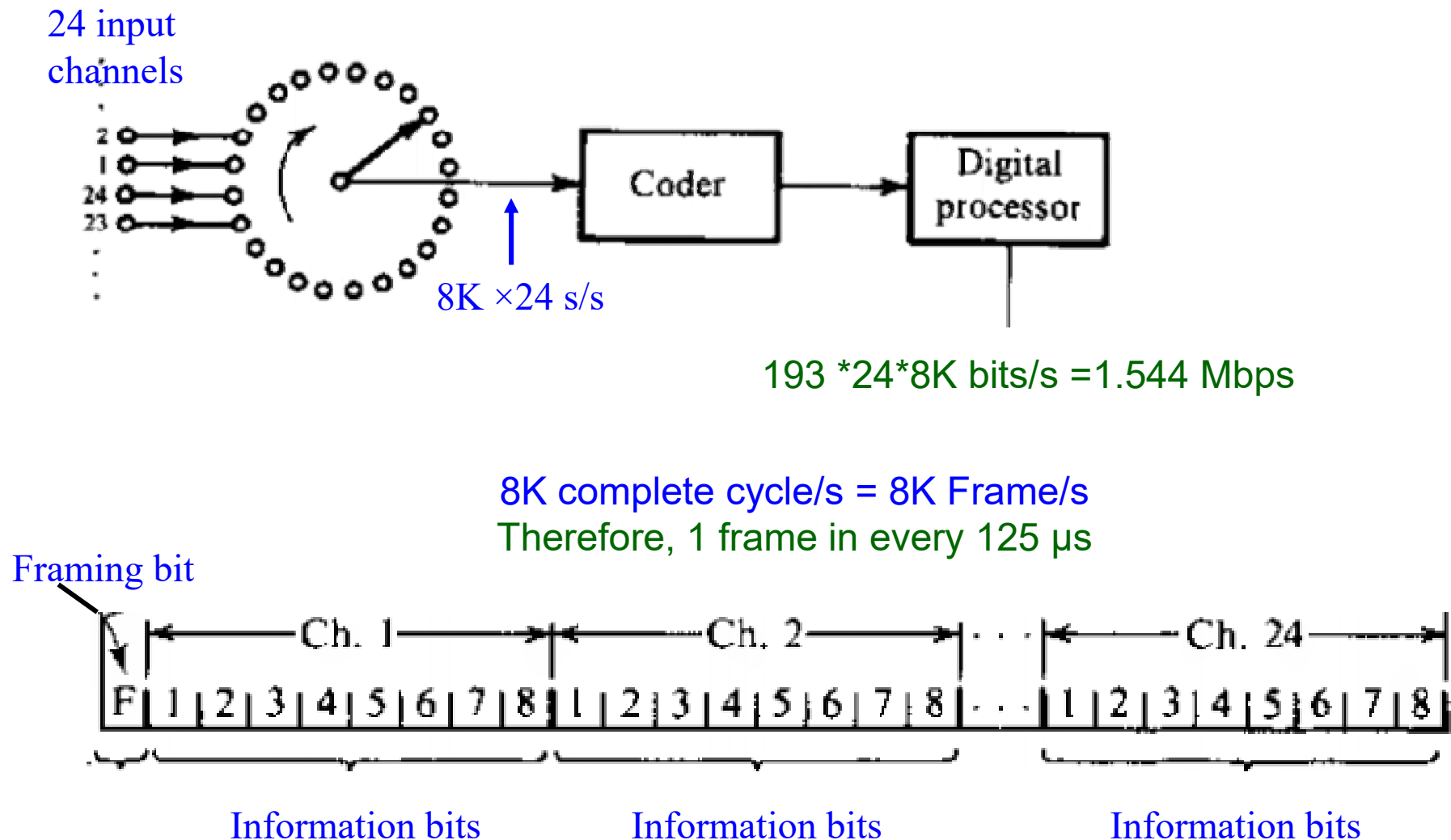
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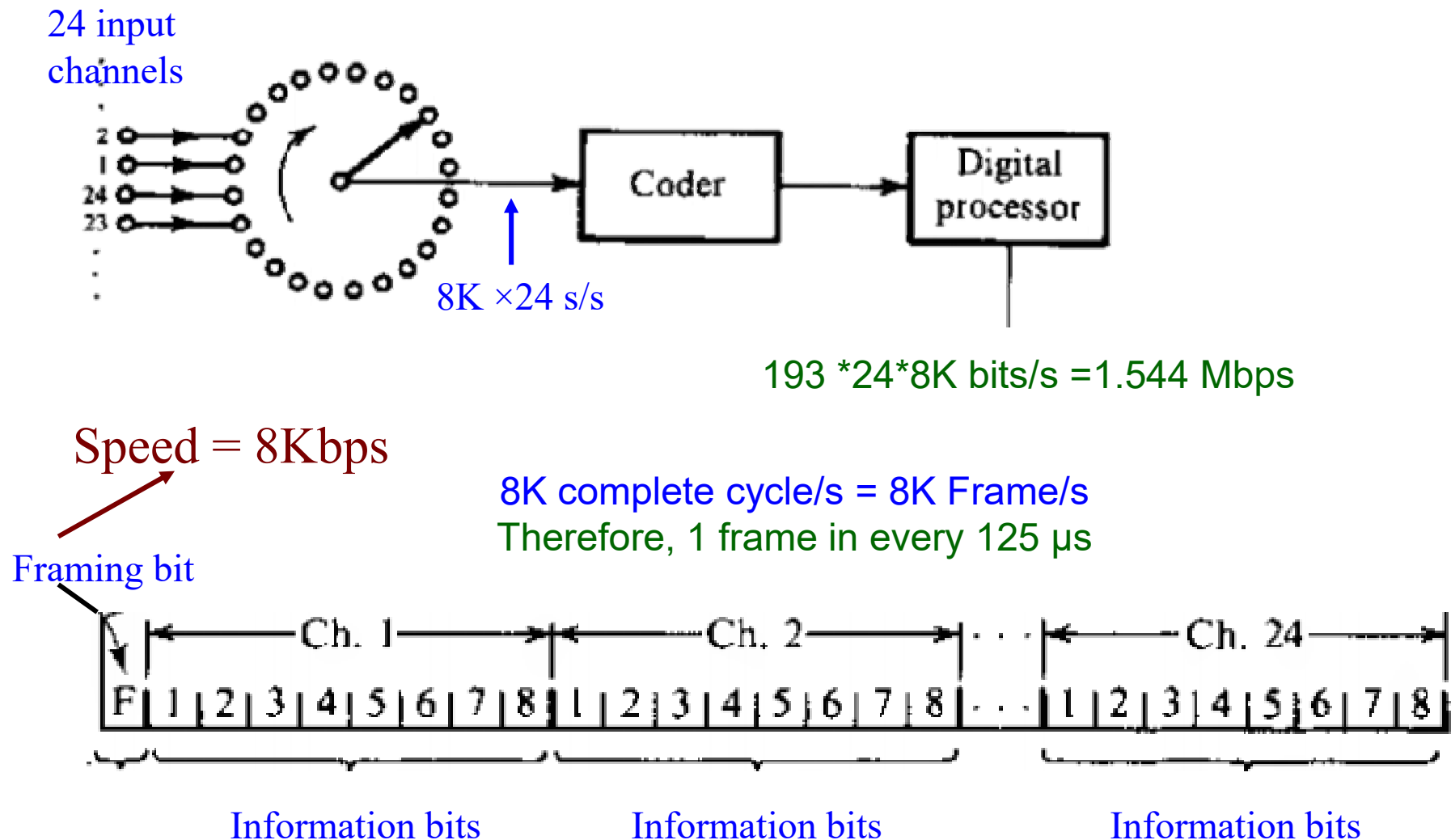
PCM in TDM: T1 Carrier System

- A Case Study



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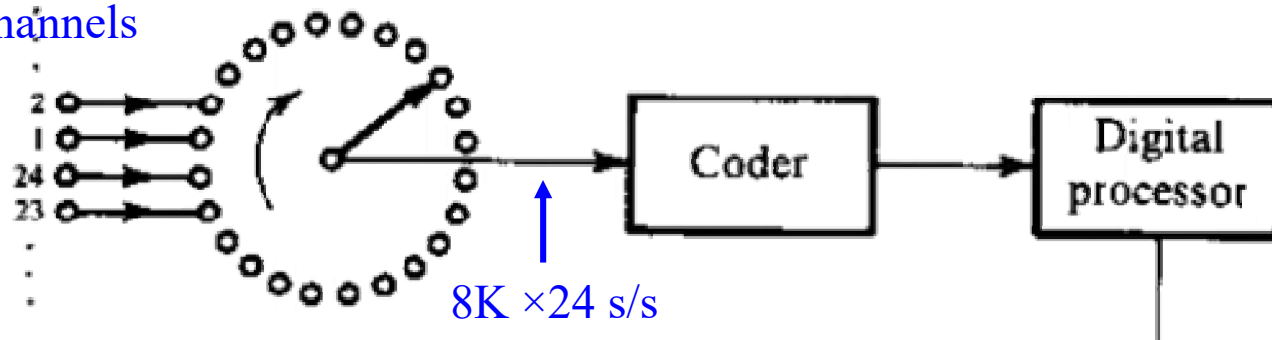
- A Case Study



PCM in TDM: T1 Carrier System

- A Case Study

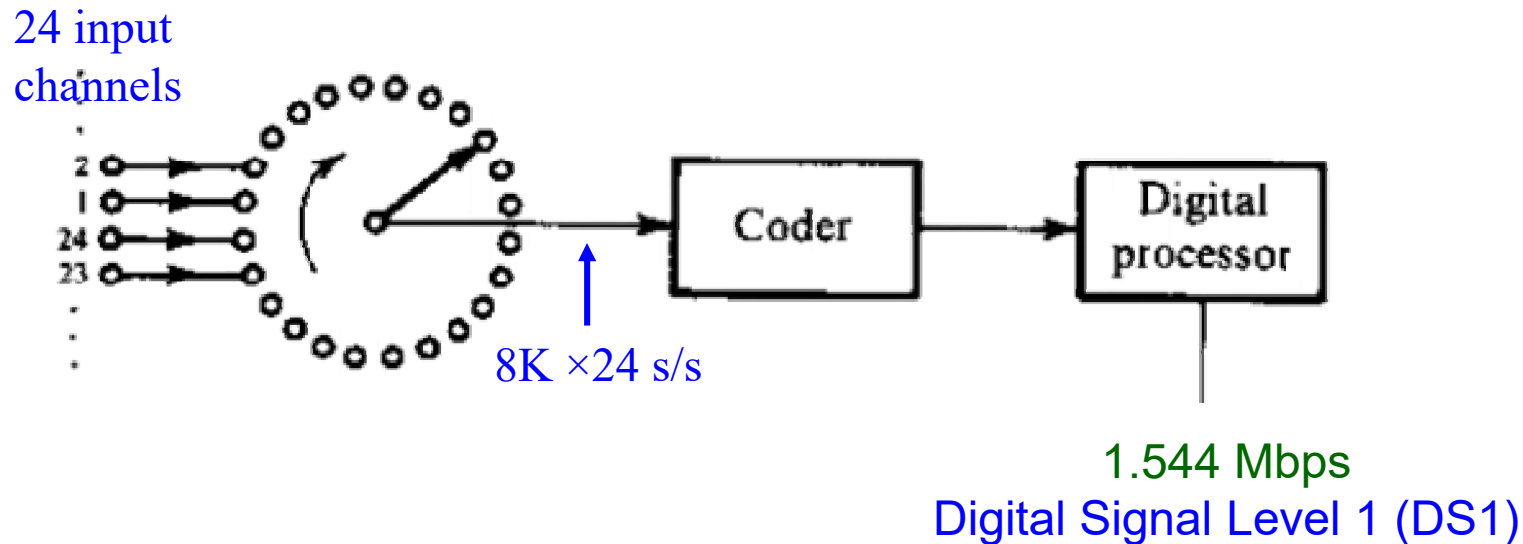
24 input
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1.544 Mbps
Digital Signal Level 1 (DS1)

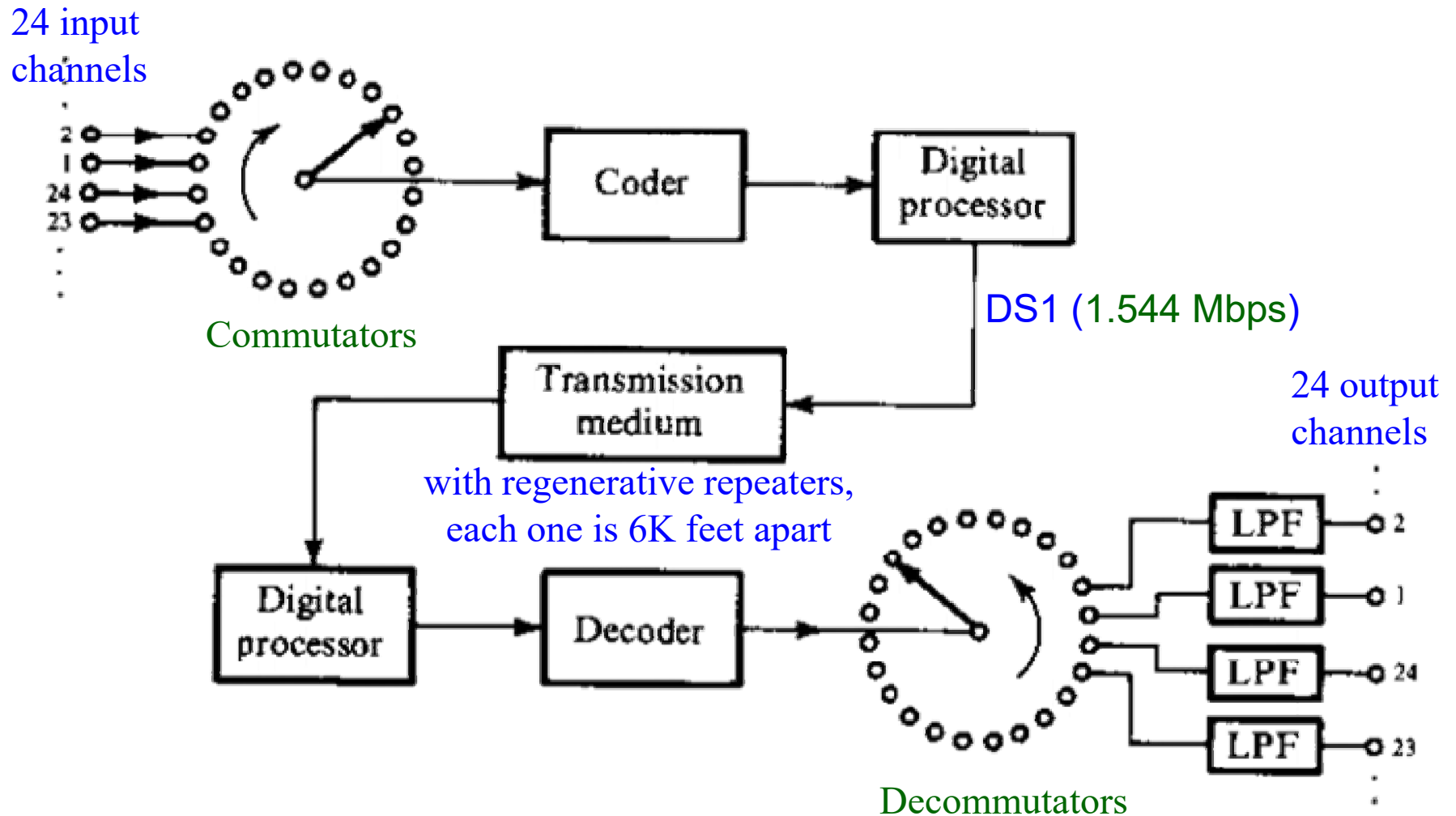
PCM in TDM: T1 Carrier System

- A Case Study

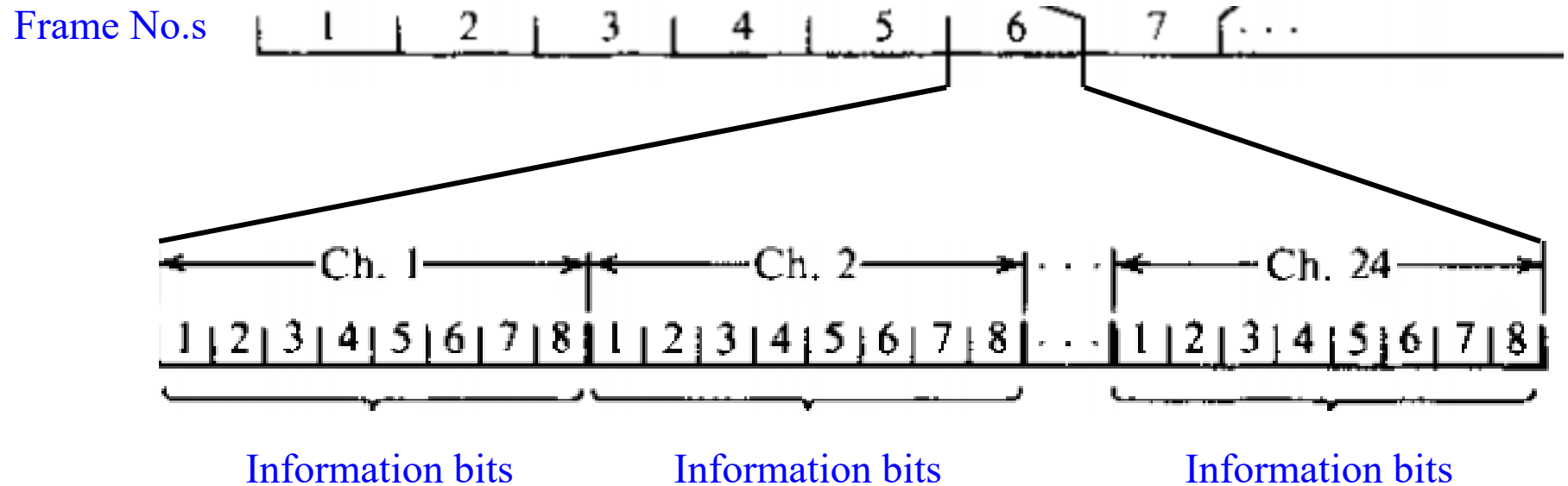


- DS1 signals can be progressively multiplexed to generate DS2, DS3, . . .
- T1 carrier system is used in US and Japan
- 30-channel multiplexed signal (2.048 Mbps) is used elsewhere in the World

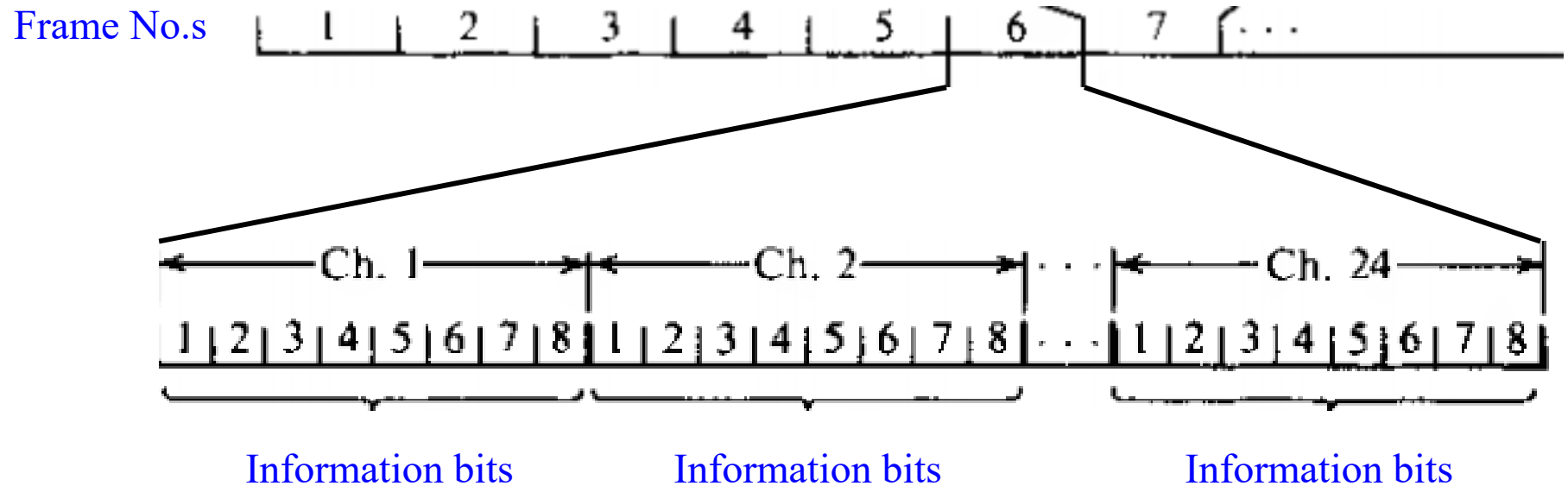
PCM in TDM: T1 Carrier System - A Case Study



Synchronizing and Signaling in T1 Carrier

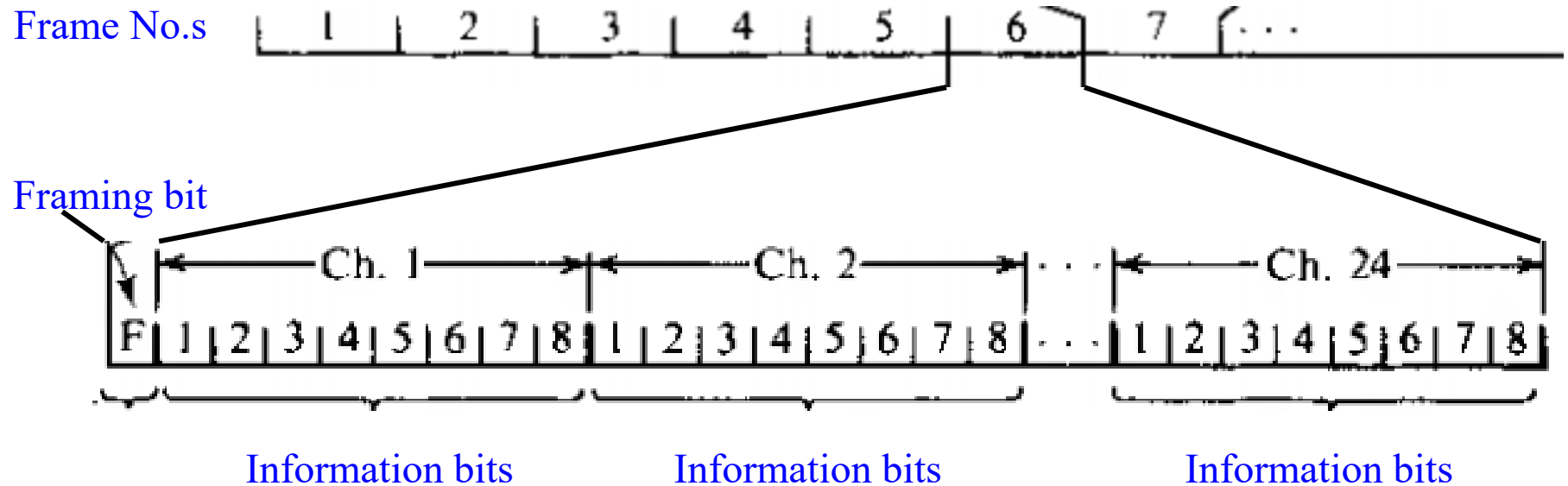


Synchronizing and Signaling in T1 Carrier



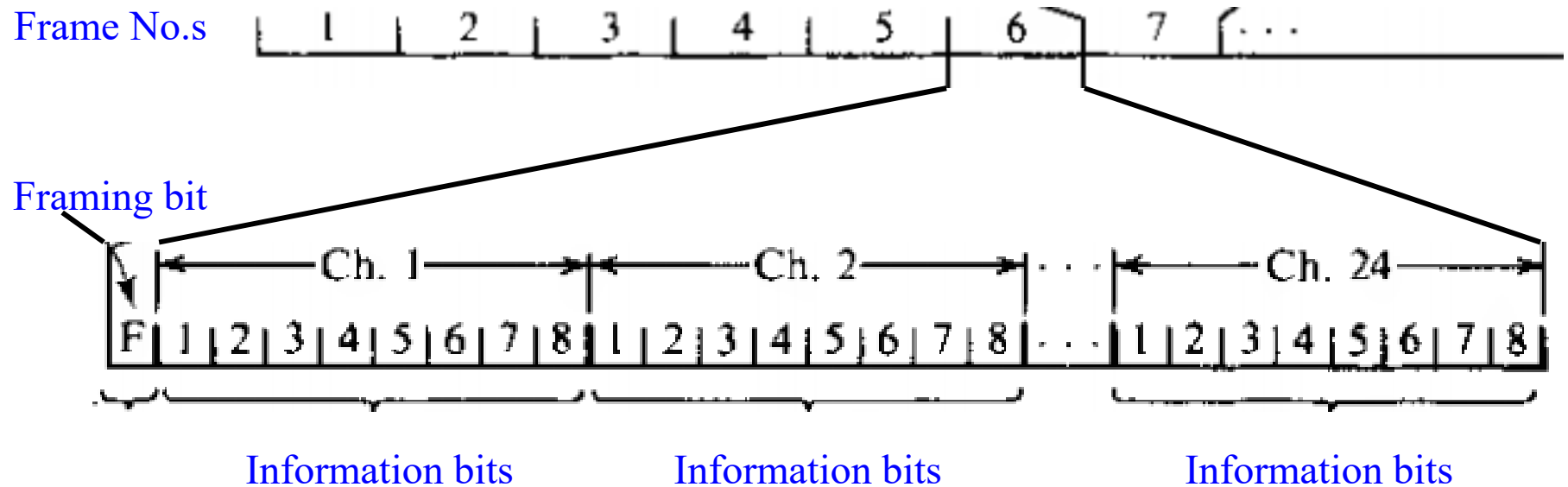
We need to know where each frame begins

Synchronizing and Signaling in T1 Carrier



A special bit (framing bit) added at the beginning of each frame
Framing bit's speed is 8 kbps

Synchronizing and Signaling in T1 Carrier

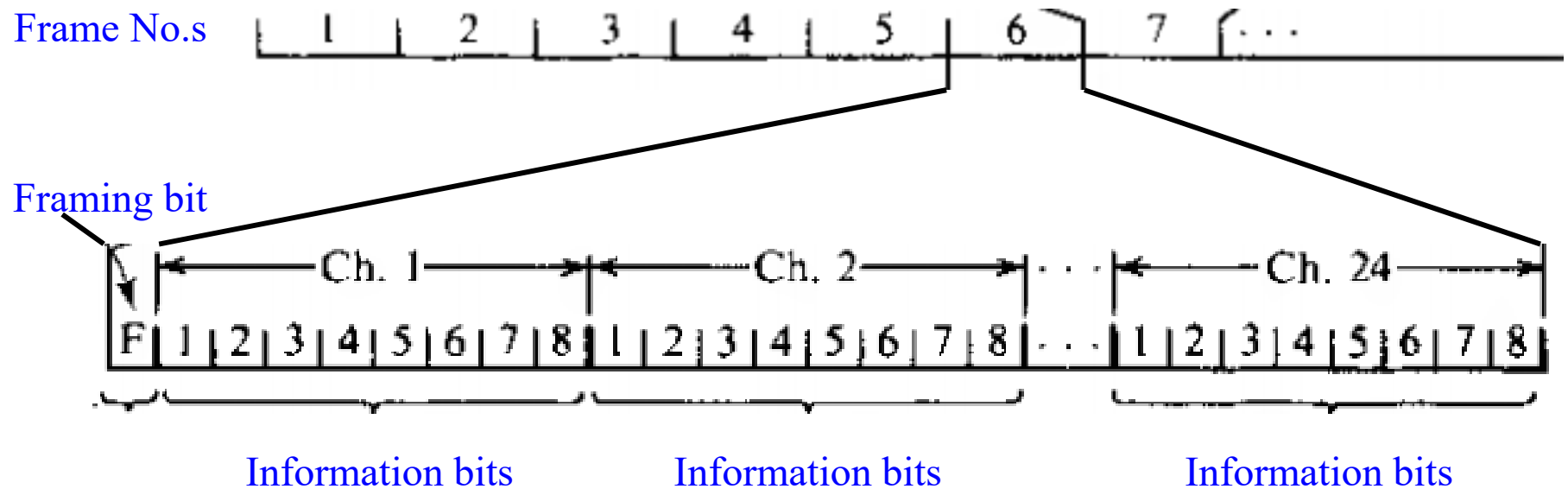


A special bit (framing bit) added at the beginning of each frame

Framing bit's speed is 8 kbps

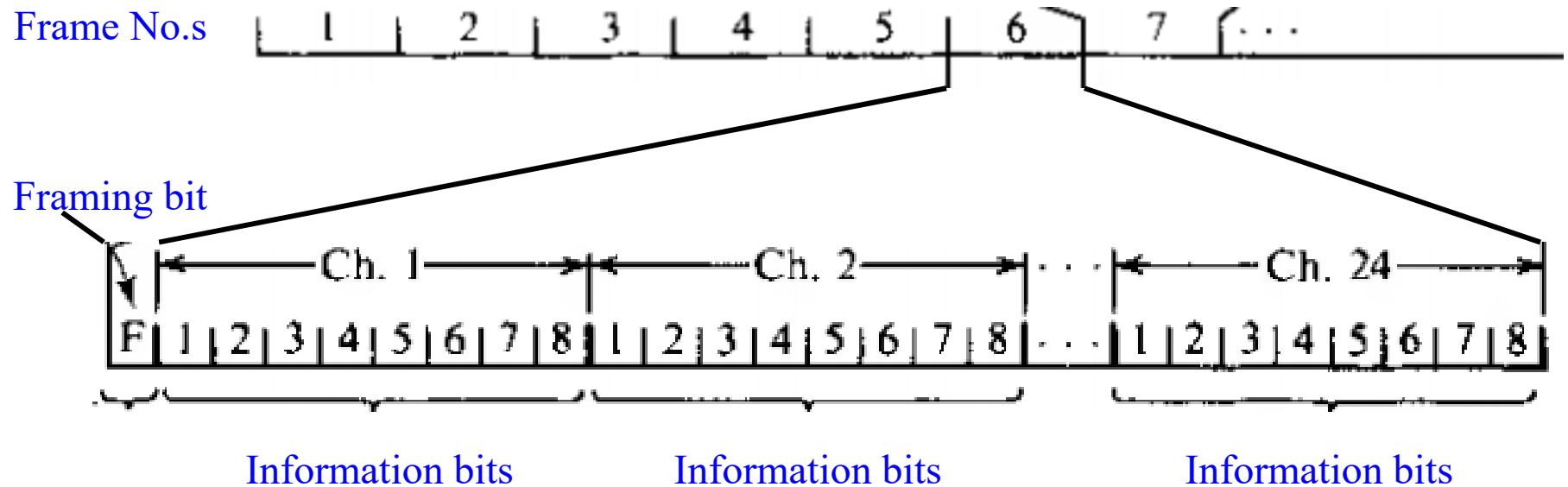
As if an extra channel (Framing channel) generates these framing bits

Synchronizing and Signaling in T1 Carrier



A special bit (framing bit) added at the beginning of each frame
Sequence of framing bits, one from each frame, form a special pattern

Synchronizing and Signaling in T1 Carrier

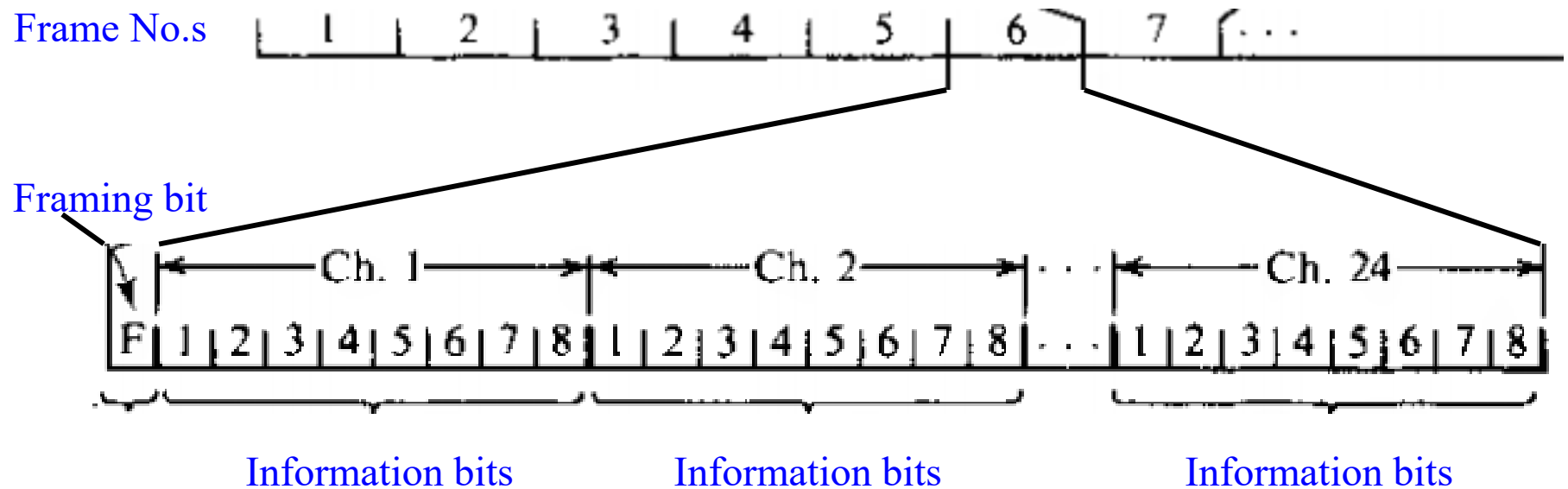


This special pattern is searched for synchronization.

Detection time: 0.4 – 6 ms

Reframe time: 50 ms

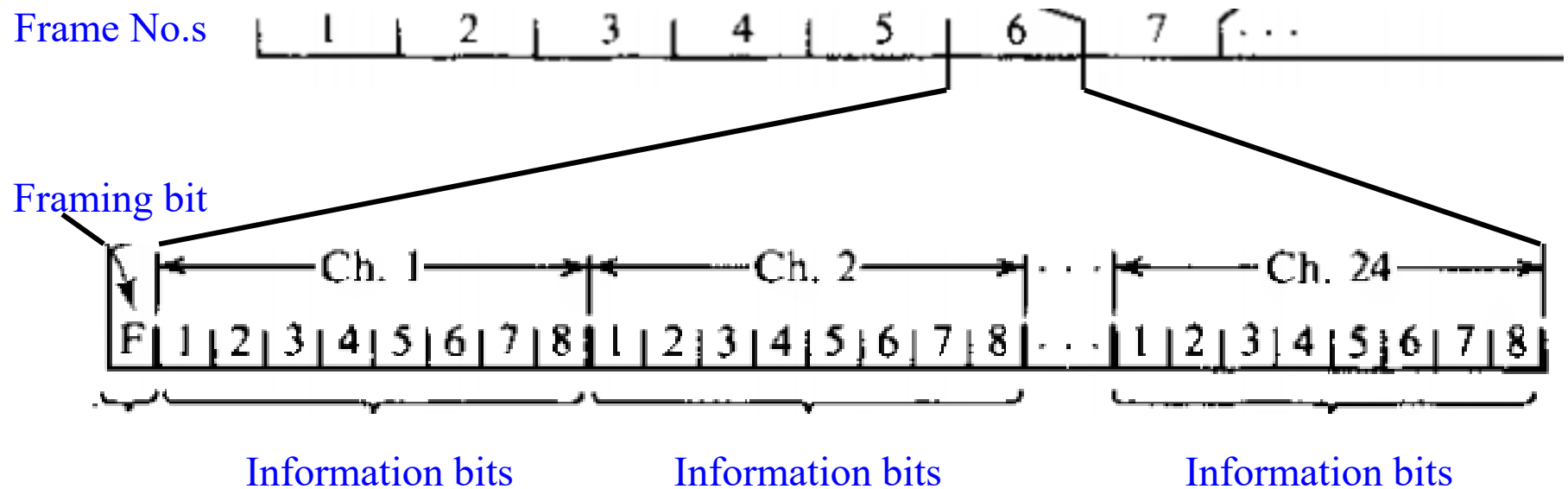
Synchronizing and Signaling in T1 Carrier



T1 requires to also transmit signaling bits

- Dialing pulses
- On-hook/off-hook signals

Synchronizing and Signaling in T1 Carrier

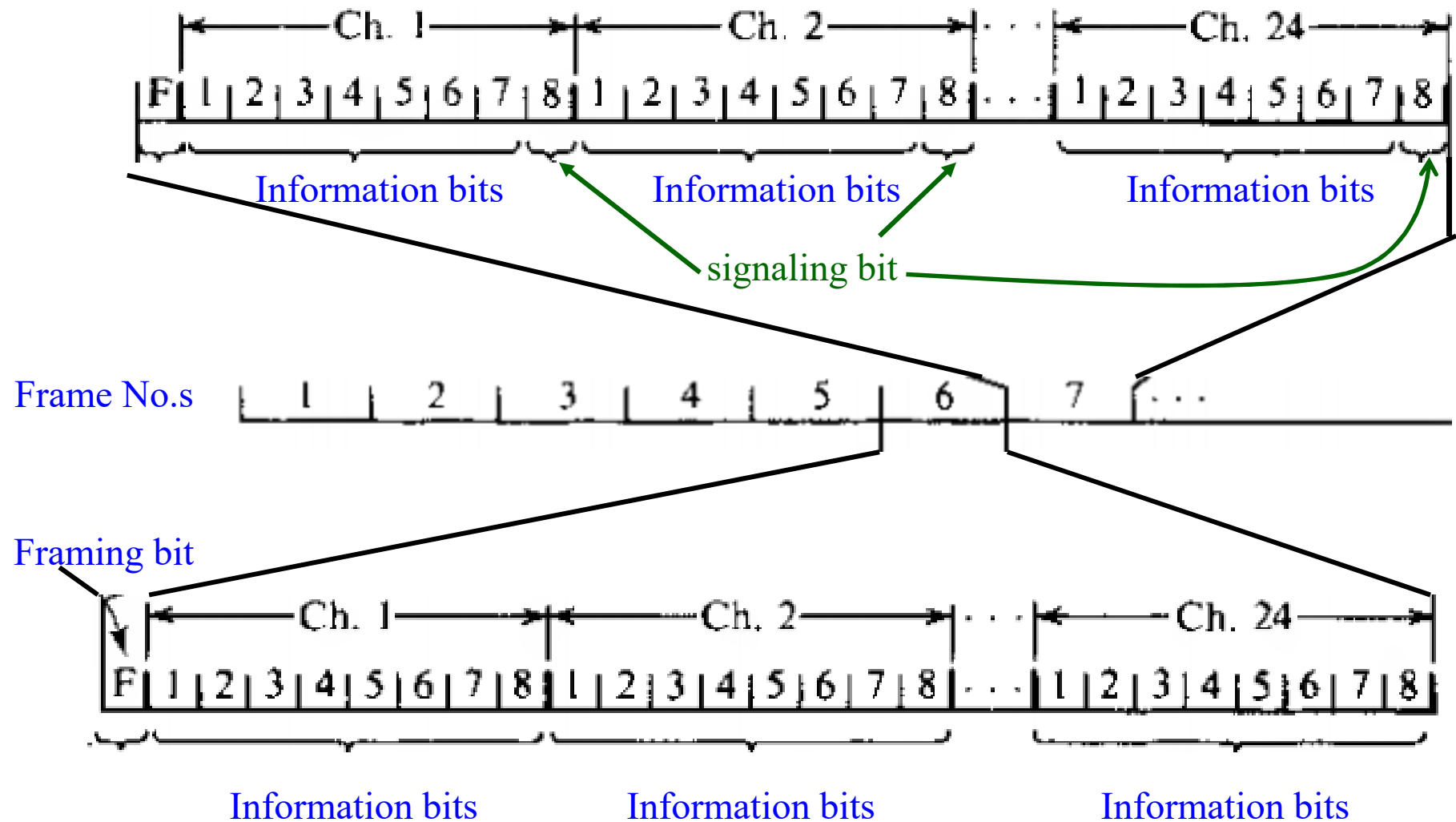


T1 requires to also transmit signaling bits

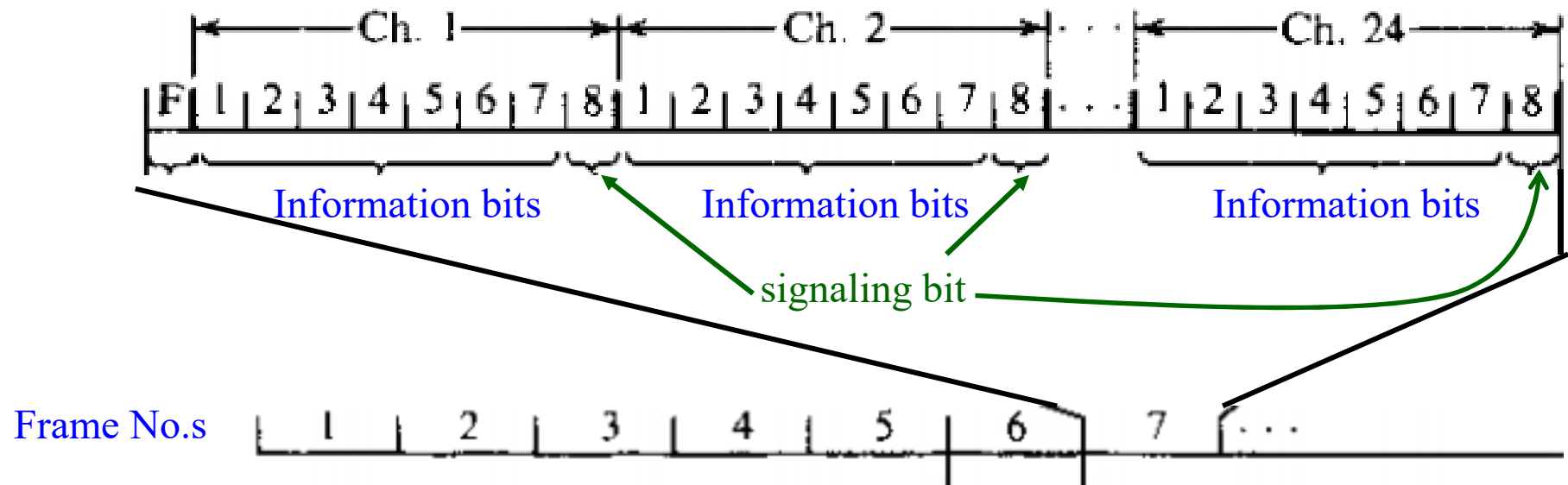
- Dialing pulses
- On-hook/off-hook signals

All 8 slots are occupied for information bits

Synchronizing and Signaling in T1 Carrier



Synchronizing and Signaling in T1 Carrier

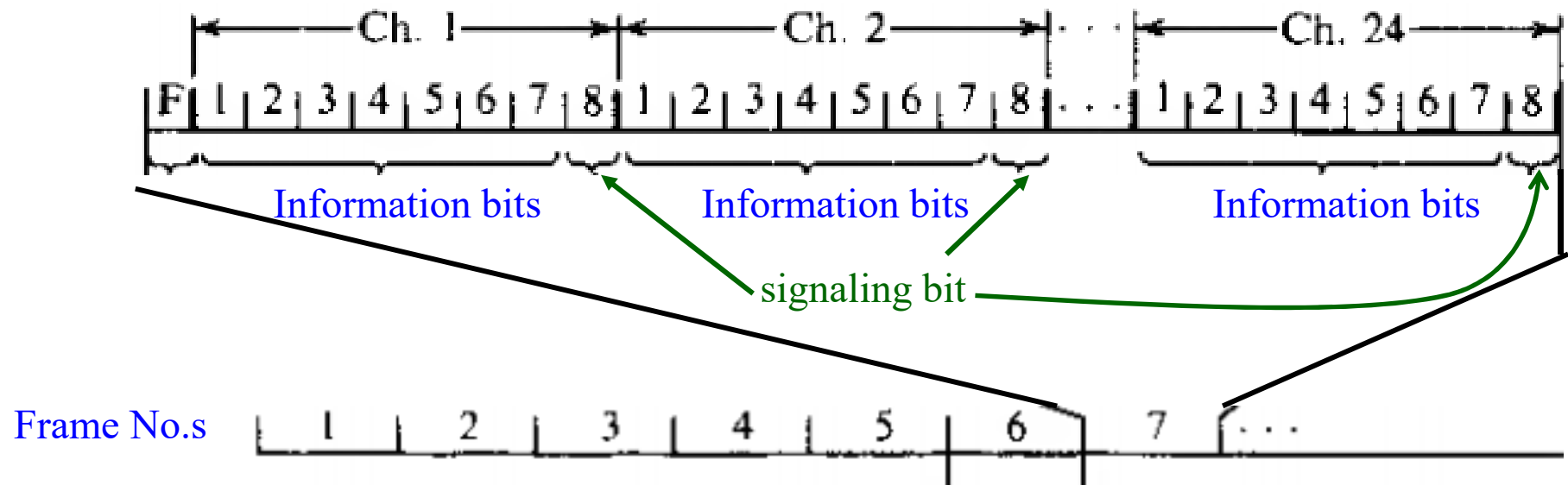


ONLY every 6th sample contains signaling bit in its LSB position

Therefore, Frame No.s 1, 7, 13 ... contains signaling bits

Signaling bit rate = $8K/6$ bps = 1333 bps

Synchronizing and Signaling in T1 Carrier

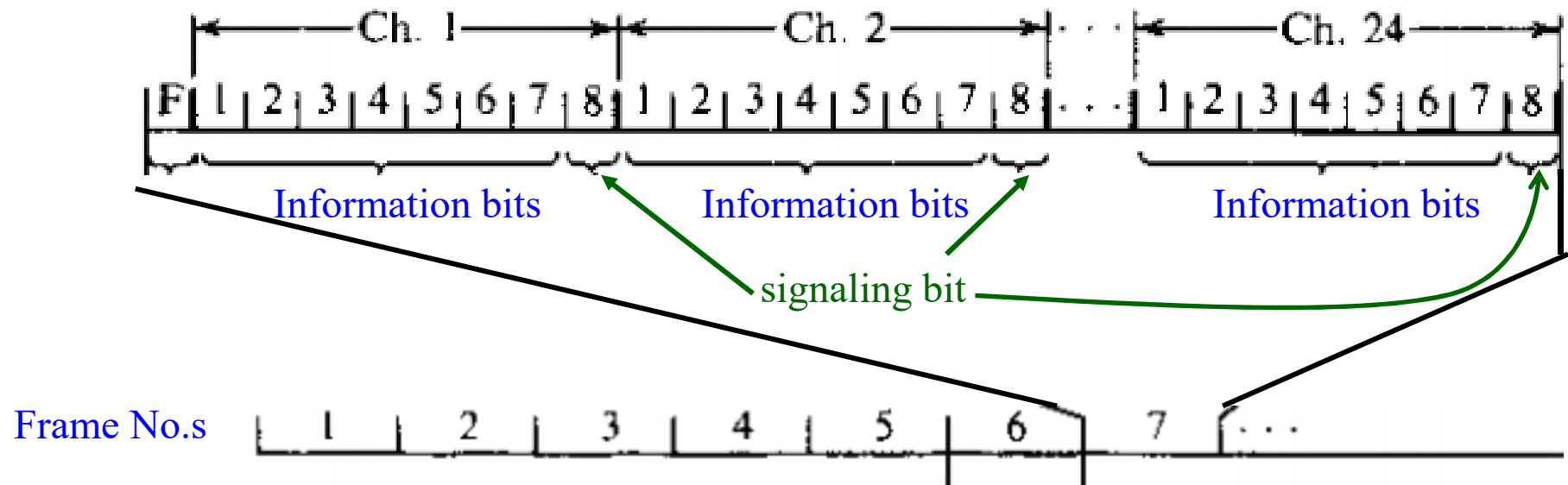


In each of Frame No.s 1, 7, 13 ...

$7 \times 24 = 168$ information bits, 24 signaling bits, 1 framing bit

Total $168 + 24 + 1 = 193$ bits

Synchronizing and Signaling in T1 Carrier



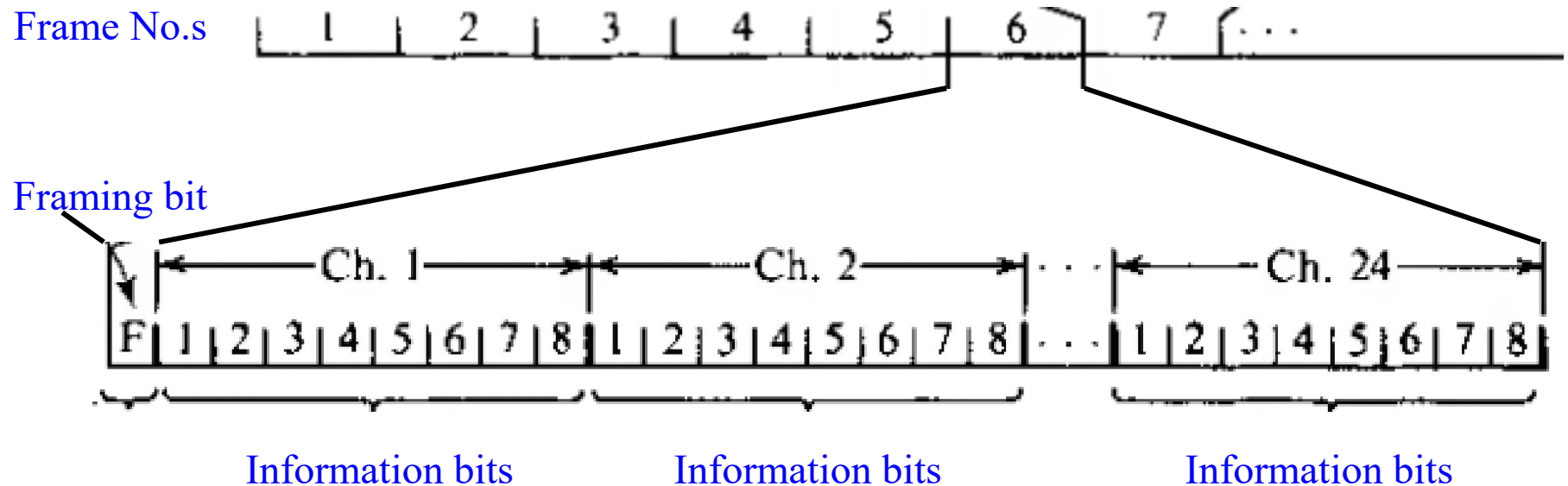
In each of Frame No.s 1, 7, 13 ...

$7 \times 24 = 168$ information bits, 24 signaling bits, 1 framing bit
 Total $168 + 24 + 1 = 193$ bits

Robbed-bit-signaling

Encoding is $7\frac{5}{6}$ encoding

Synchronizing and Signaling in T1 Carrier

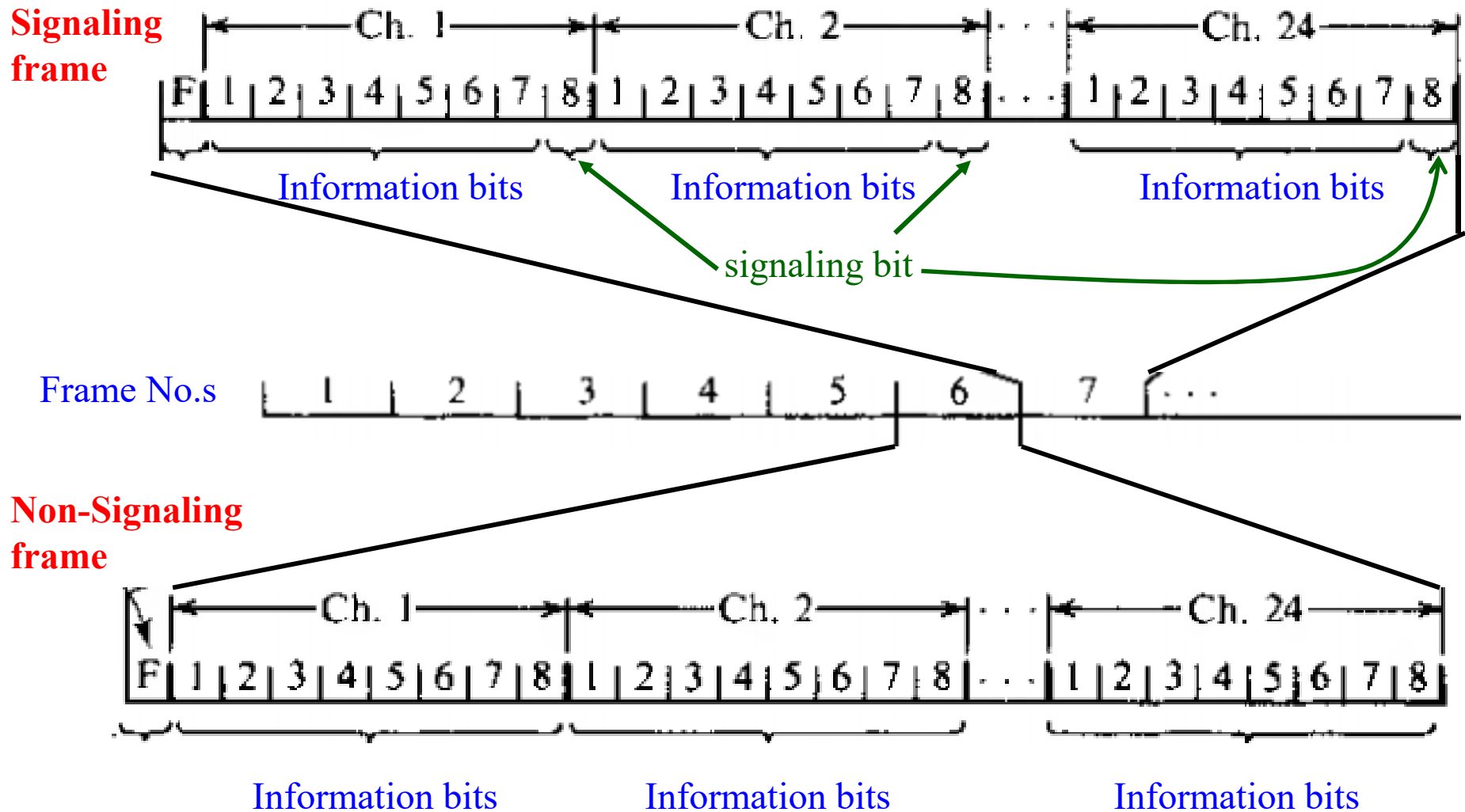


In all other frames **except** Frame No.s 1, 7, 13 ...

$8 \times 24 = 192$ information bits, 1 framing bit

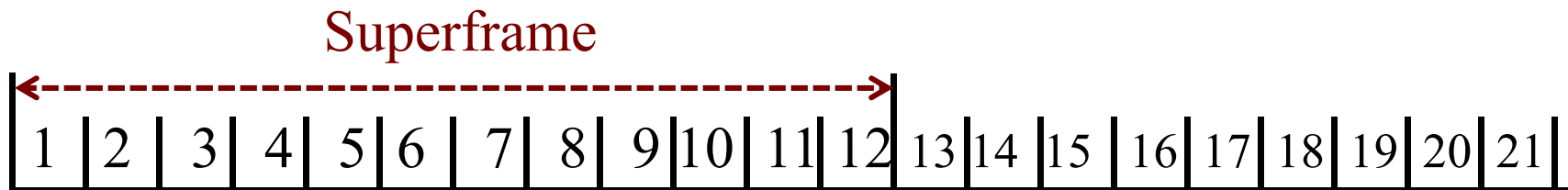
Total $192 + 1 = 193$ bits

How to Differentiate Signaling and Non-Signaling Frames?



How to Differentiate Signaling and Non-Signaling Frames?

Concept of Superframe

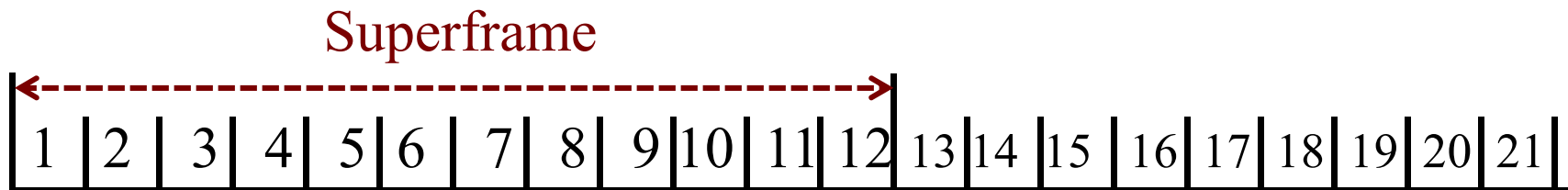


Frame No.s

- 12 frames make a Superframe
- Framing bits form a special pattern 100111011100

How to Differentiate Signaling and Non-Signaling Frames?

Concept of Superframe

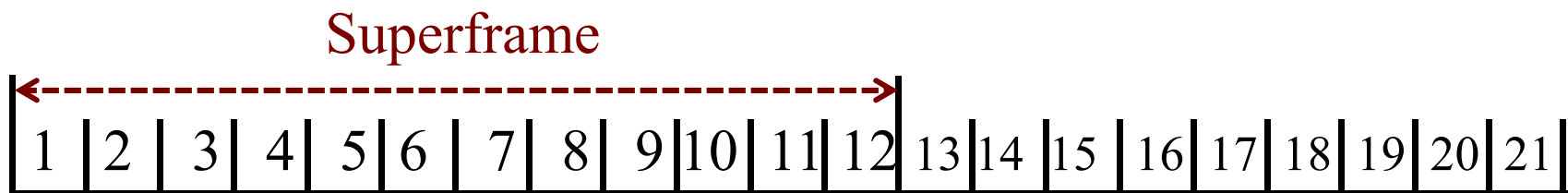


Frame No.s

- The special pattern **100111011100** repeats in every 12 frames
- Identification of 6th and 12th frame is easy, and so is signaling frames.

How to Differentiate Signaling and Non-Signaling Frames?

Concept of Superframe



Frame No.s

- 2 signaling frames in a superframe
- 2 signaling bits in a superframe
- 4 state signaling is possible